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sense and simplicity

DICOM

in a nutshell

Kishan Harwalkar
ICAP System Department
May 03, 2013

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Agenda

- Introduction to DICOM
- Information Model
- Data Encoding
- Network Model
- Image & Presentation State
- DICOM Conformance

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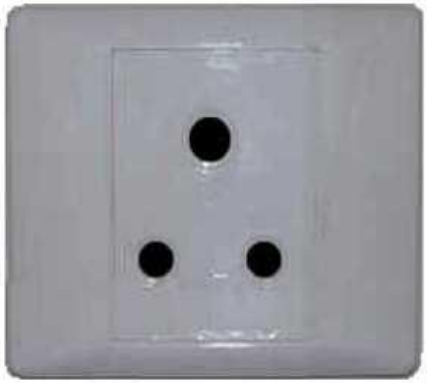

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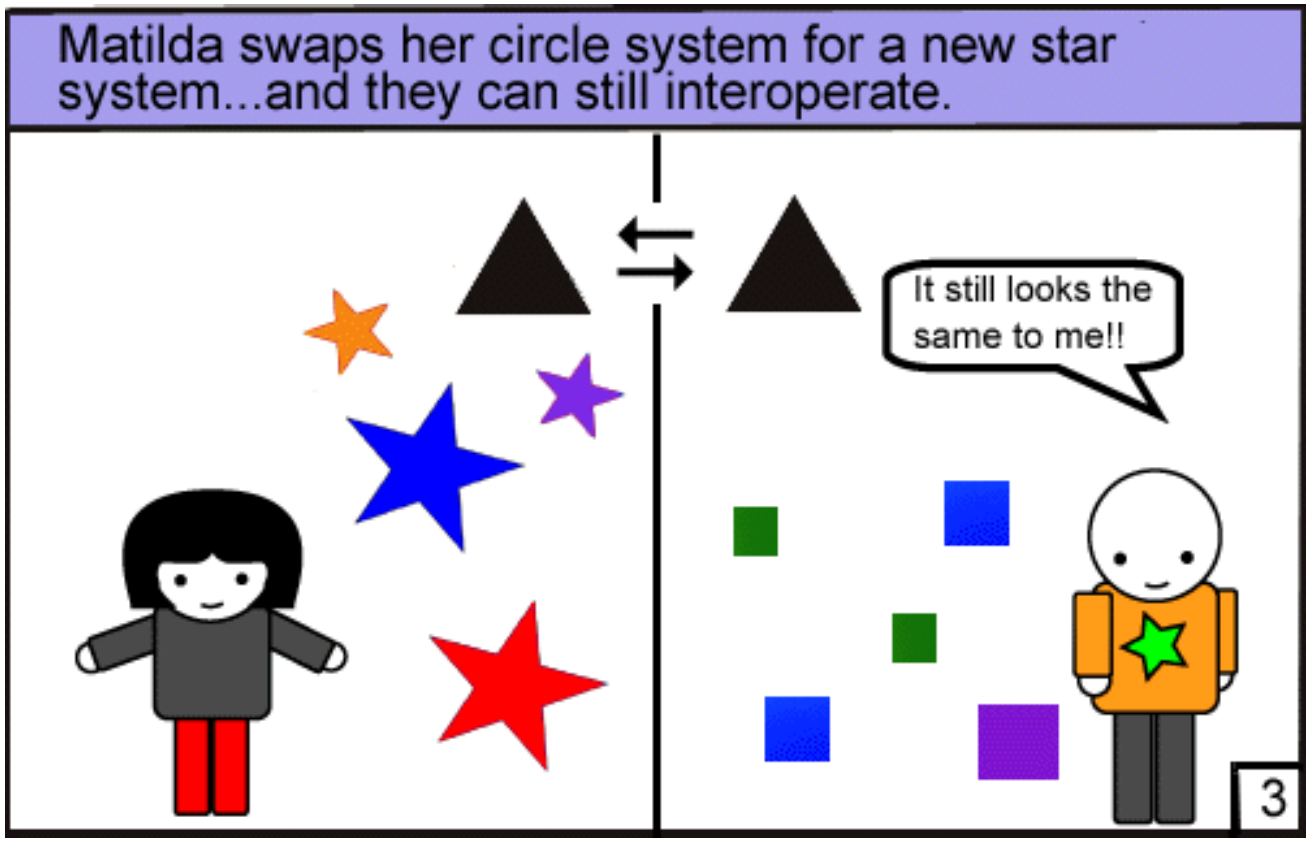
Communication Story



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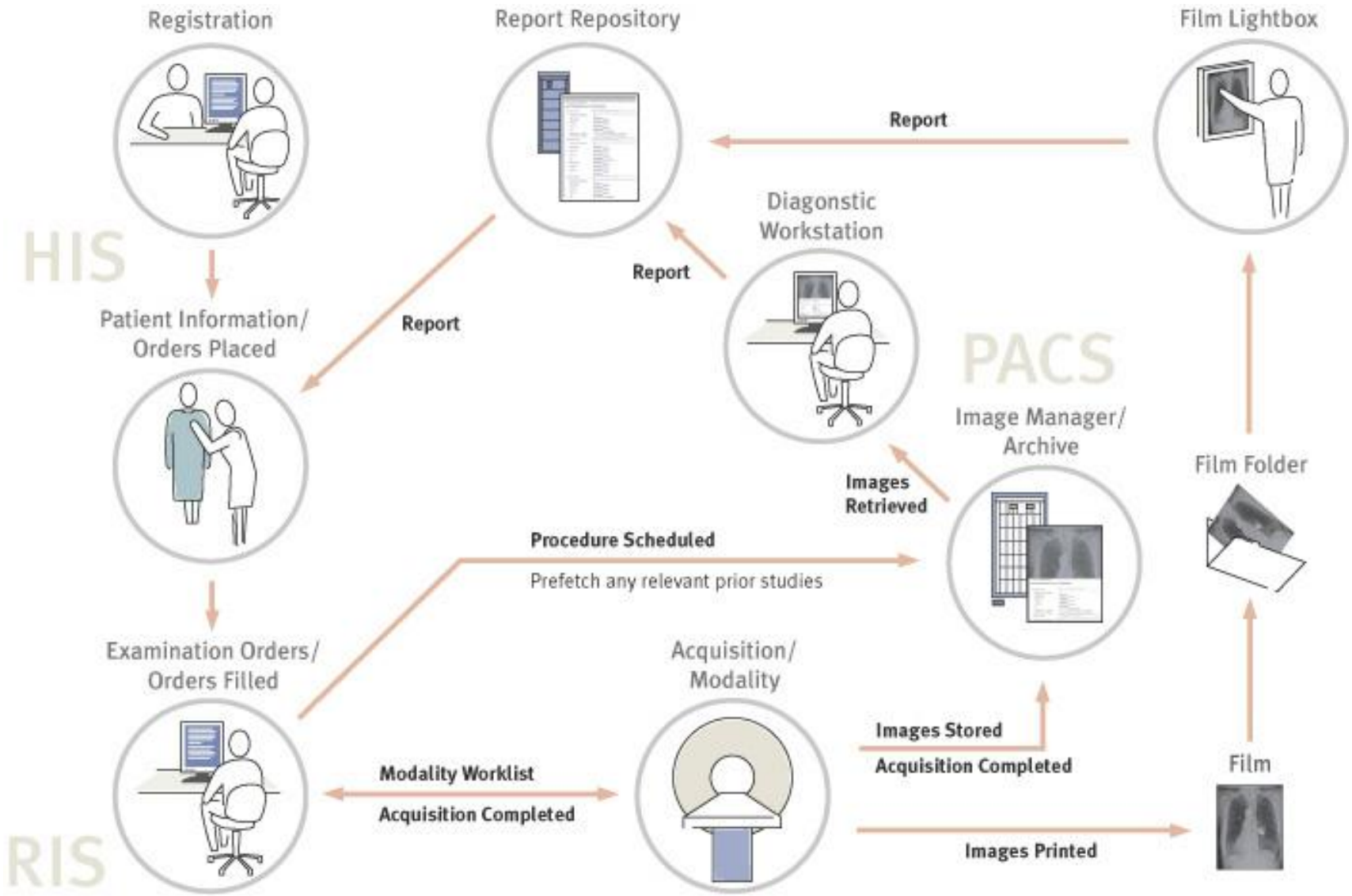
Healthcare Workflow



Standards for Healthcare

Health Level Seven®

DICOM℠

Digital Imaging and Communications in Medicine

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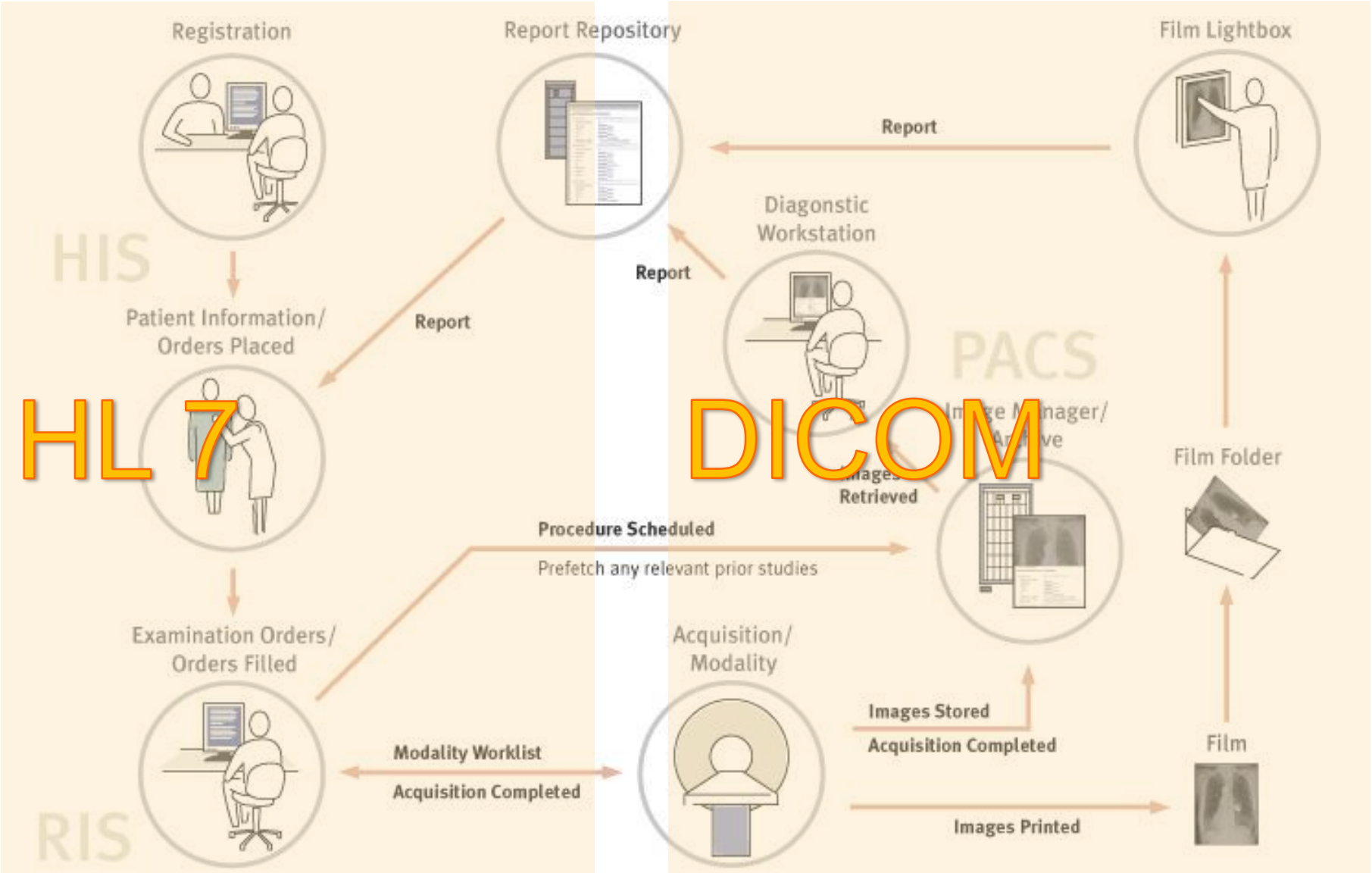
Healthcare Standards



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History of DICOM

- SPI – Early 1980
- ACR-NEMA 1.0 in 1985
- ACR-NEMA 2.0 in 1988
- DICOM 3.0 in 1992-1993
- Continuous development since then – DICOM 2012 is out

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What is DICOM

- Standard for Medical Images and associated data.
  - Defines storage and communication model for Medical Images & associated data
    - A Set of protocols to be followed by devices claiming conformance to the standard
    - Semantics & Syntax of commands and associated information which can be exchanged
    - Conformance Requirements for Implementations

- DICOM Additions
  - Released as supplements to the existing standard
    - Semantically same as the existing specifications and driven by industry needs

Stakeholders

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NEMA
Setting Standards for Excellence

GE Healthcare

AGFA HealthCare

MIT MEDICAL IMAGING & TECHNOLOGY ALLIANCE
A DIVISION OF NEMA

JOHNS HOPKINS MEDICINE

SIEMENS medical

MERGE

AMERICAN COLLEGE OF CARDIOLOGY

EUROPEAN SOCIETY OF CARDIOLOGY®

SIIM

acr American College of Radiology

HIMSS The Source for Healthcare Information

SFR Société Française de Radiologie

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Attribute

- Smallest piece of information in DICOM

- Anatomy of an Attribute
  - Name
  - Tag
  - Value Representation
  - Value Multiplicity
  - Value

Attribute

- Tag
  - Hexadecimal Number
    - Group gggg
    - Element eeee
  - Group
    - Even – DICOM Defined
    - Odd – Vendor Defined
  - Example
    - 0010 0010 – Patient Name
    - 0010 0020 – Patient ID
    - 0010 0030 – Patient Birth Date

Attribute

- Value Representation (VR)
  - Data Type and the Format
  - Length
  - Permissible Characters
  - Categories
    - Mathemetical – FL, SS, DS etc
    - Character String – CS, ST etc
    - Special Characteristics – DA, DT etc
    - Other – OB, OW and SQ

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Attribute: Value Representation

VR Name	Definition	Character Repertoire	Length of Value
CS Code String	A string of characters with leading or trailing spaces (20H) being non-significant.	Uppercase characters, "0"-"9", the SPACE character, and underscore "_", of the Default Character Repertoire	16 bytes maximum
FL Floating Point Single	Single precision binary floating point number represented in IEEE 754:1985 32-bit Floating Point Number Format.	not applicable	4 bytes fixed
IS Integer String	A string of characters representing an Integer in base-10 (decimal), shall contain only the characters 0 - 9, with an optional leading "+" or "-". It may be padded with leading and/or trailing spaces. Embedded spaces are not allowed. The integer, n, represented shall be in the range: $-2^{31} <= n <= (2^{31} - 1)$.	"0"-"9", "+", "-" of Default Character Repertoire	12 bytes maximum

Attribute

- Value Multiplicity
  - Number of values for an Attribute
    - Example:
      - VM = 1
      - VM = 1-n
  - “\” is the Delimiter for Character Strings

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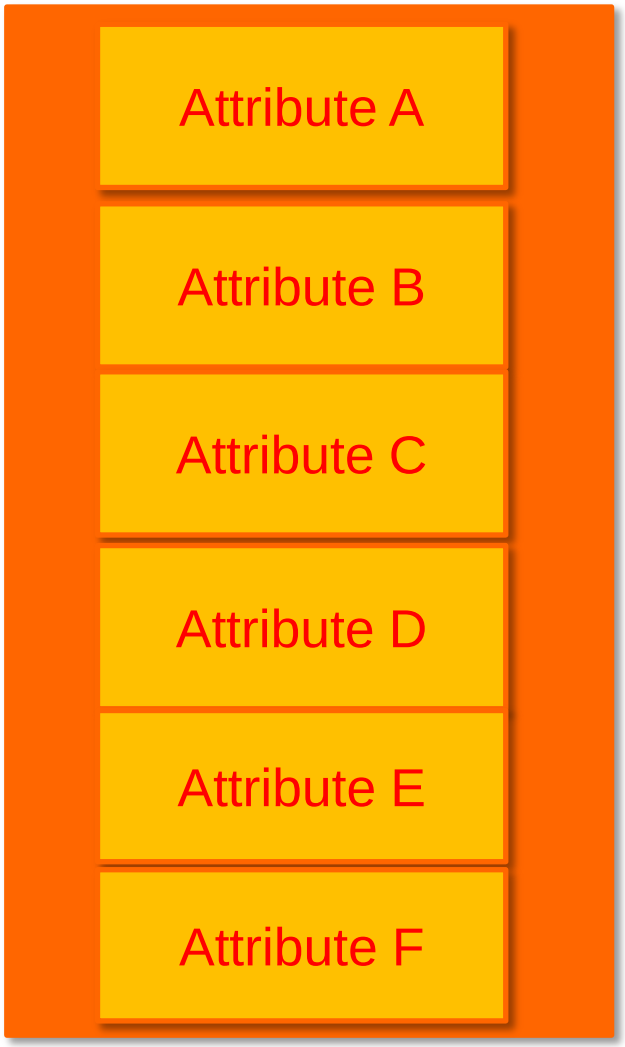
Information Object Module

- A set of related attributes are grouped into a module
- Attribute present in a module has Mandatoriness
- Mandatoriness of Attribute
  - 1 Required and non null
  - 1C Required based on a condition
  - 2 Required can be null
  - 2C Required based on a condition
  - 3 Optional



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Attribute Mandatoriness

Attribute Name	Tag	Type	Attribute Description
Modality	(0008,0060)	1	Type of equipment that originally acquired the data used to create the images in this Series. See C.7.3.1.1.1 for Defined Terms.
Series Instance UID	(0020,000E)	1	Unique identifier of the Series.
Series Number	(0020,0011)	2	A number that identifies this Series.
Laterality	(0020,0060)	2C	Laterality of (paired) body part examined. Required if the body part examined is a paired structure and Image Laterality (0020,0062) or Frame Laterality
Series Date	(0008,0021)	3	Date the Series started.
Series Time	(0008,0031)	3	Time the Series started.
Performing Physicians' Name	(0008,1050)	3	Name of the physician(s) administering the Series.

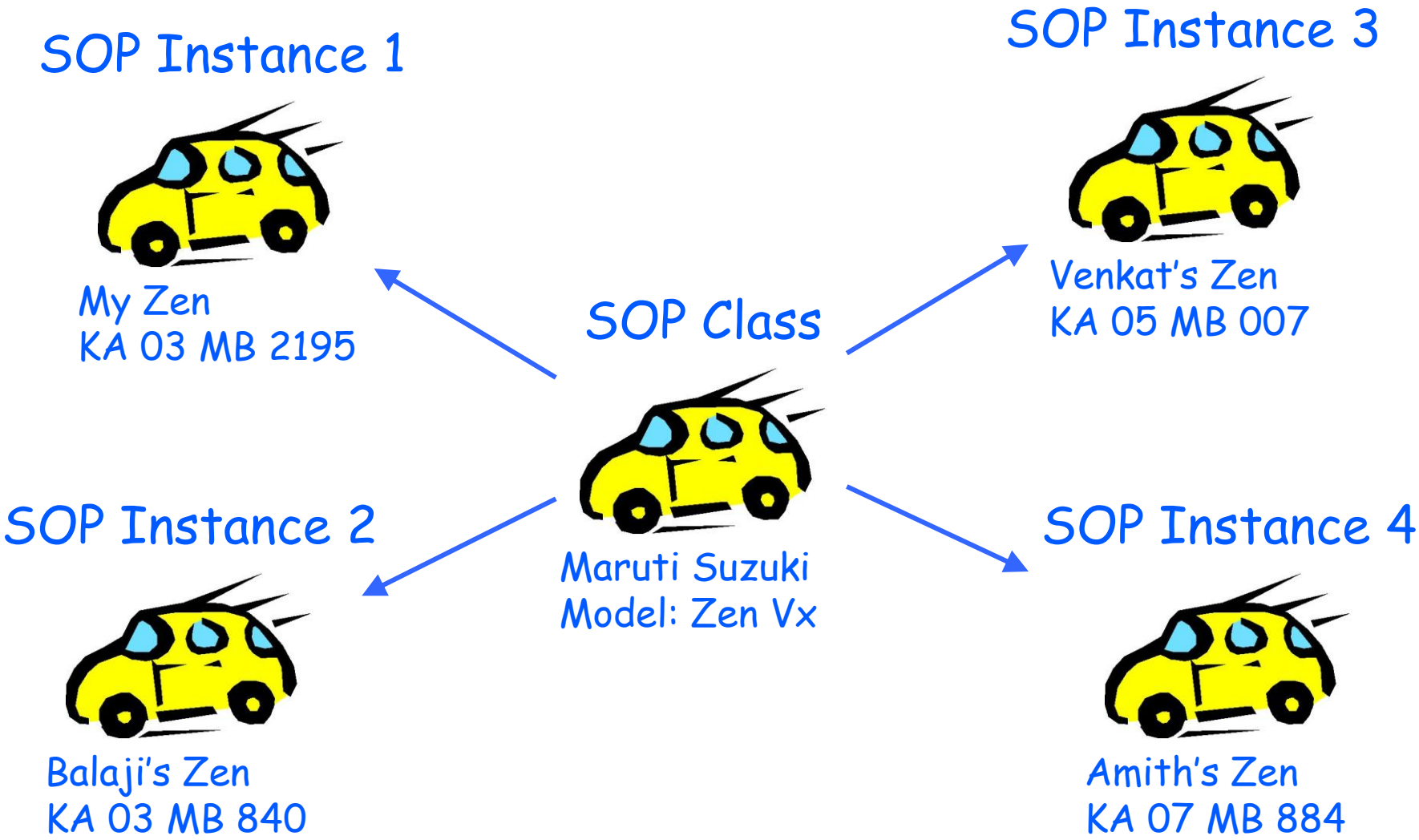
Information Object Definition (IOD)

- Collection of attributes representing an Object
- IODs are Identified by SOP Class
- SOP Class
  - SOP Class = Service + Object (IOD)
  - Defines the Type of information Stored along with Service associated (e.g: MR Image Storage)
  - Identified by an Unique Identifier called SOPClassUID
  - At least one SOP Class for each Modality (MR/CT/US)

SOP Instance

- Instance of a SOP Class
- Similar to OO Concepts Class/Instance
- Each Instance is identified by an Unique Identifier called SOP Instance UID

Information Object Definition (IOD)



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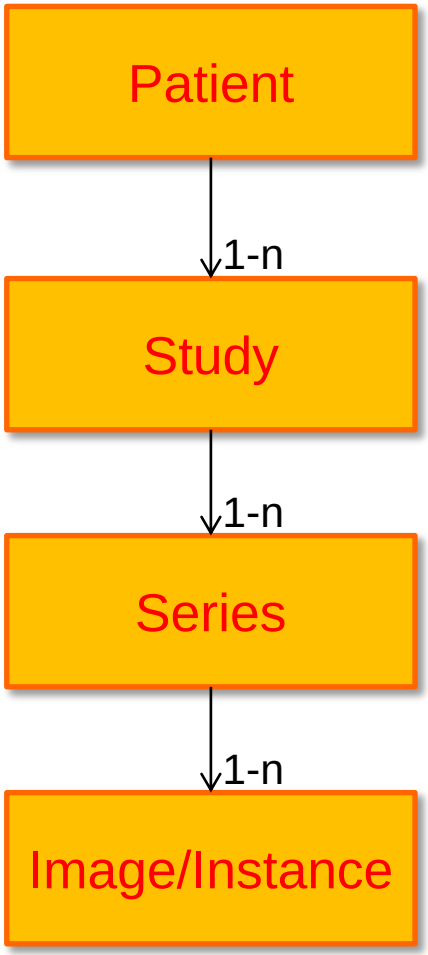
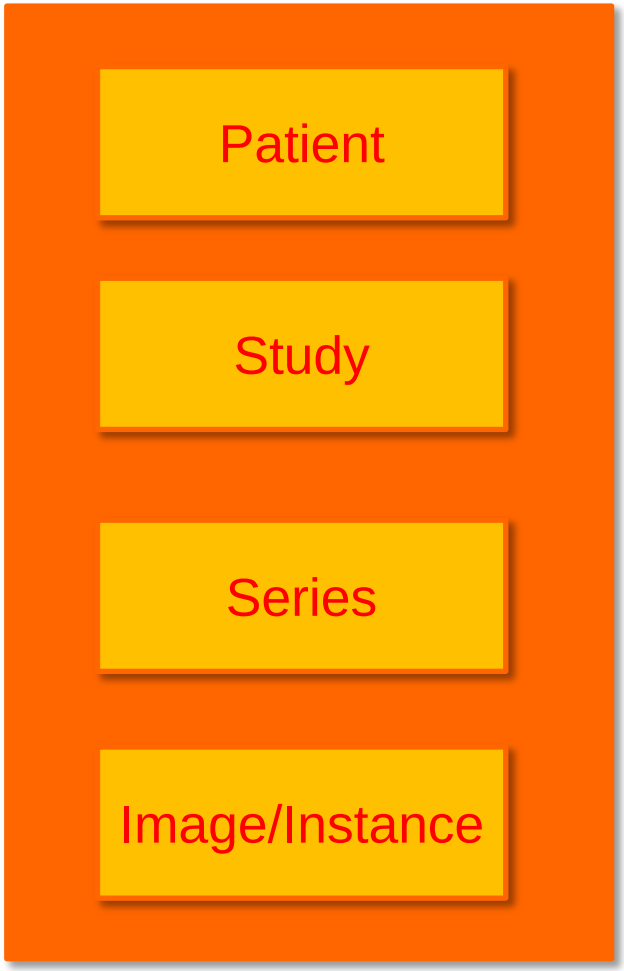
DICOM Object Models



Normalized Model



Composite Model

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Computed Radiography Image IOD

IE	Module
Patient	Patient
Study	General Study
	Patient Study
Series	General Series
	CR Series
Equipment	General Equipment
Image	General Image
	Image Pixel
	Contrast/bolus
	CR Image
	Overlay Plane
	Curve
	Modality LUT
	VOILUT
	SOP Common

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Data Dictionary

CR IOD

IE	Module	Attribute

CT IOD

IE	Module	Attribute

MR IOD

IE	Module	Attribute

XA IOD

IE	Module	Attribute

SC IOD

IE	Module	Attribute

US IOD

IE	Module	Attribute

PET IOD

IE	Module	Attribute

SPECT IOD

IE	Module	Attribute

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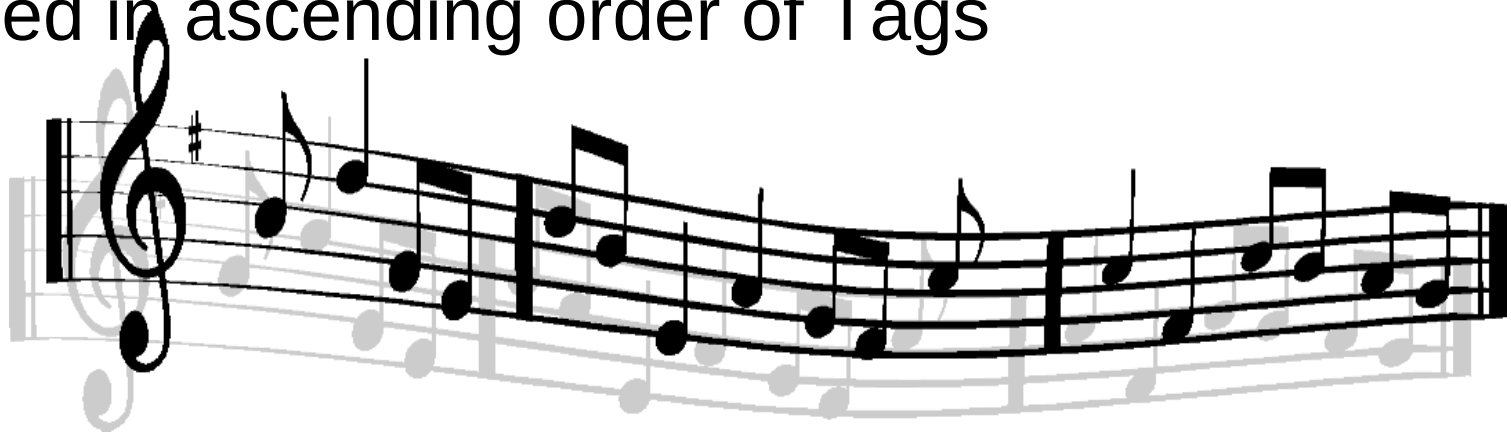
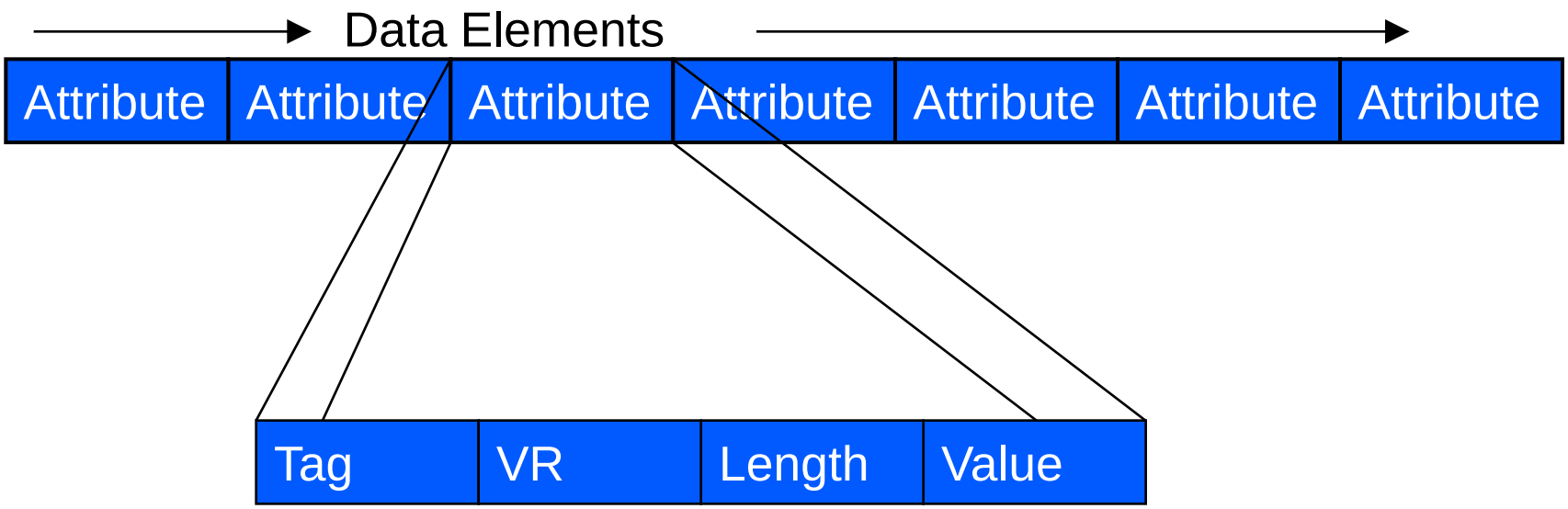
Data Set

- Structural representation of Data
  - e.g: IOD
- Sequence of Attributes
- Encoded in ascending order of Tags

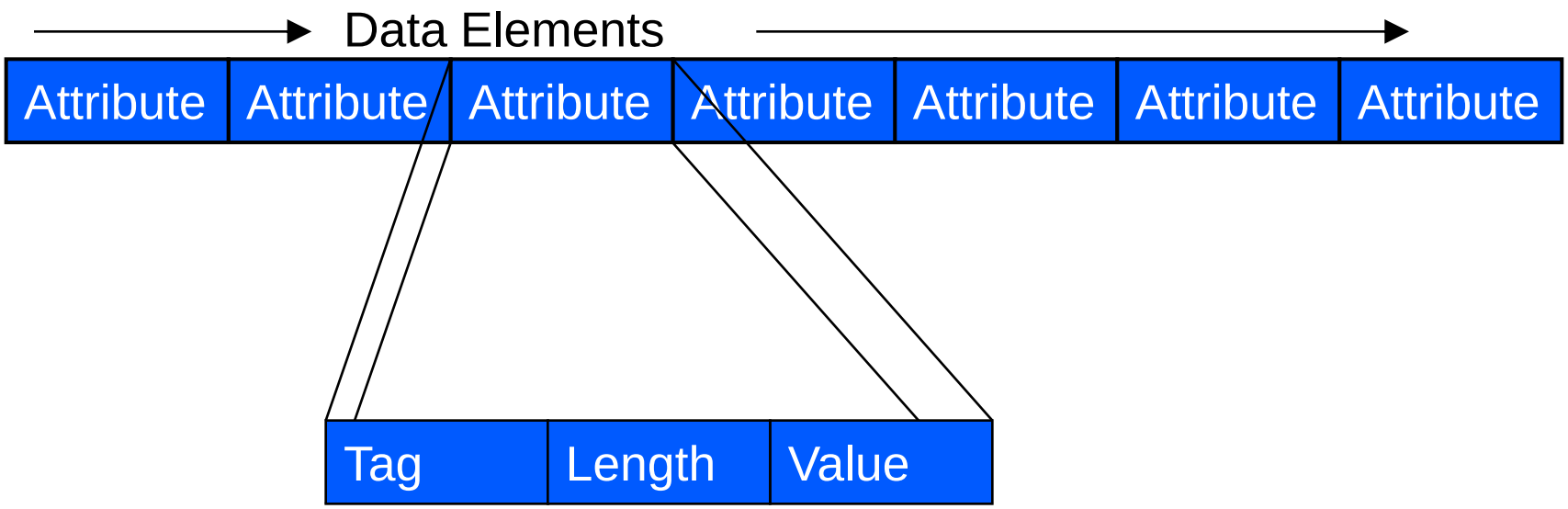


Explicit Encoding



Explicit VR

Implicit Encoding



Implicit VR

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Byte Odering

- Intel way of Encoding
  - Also known as Little Endian
    - Computer Words are stored such that Least Significant Bytes comes first

عيشي بلادي عاش اتحاد إماراتنا
عشت لشعب دينه الإسلام هديه القرآن
حصنتك بسم الله يا وطن
دام الأمان وعاش العلم يا إماراتنا
رمز العروبة كلنا نفديكِ
بالدما نرويكِ
نفديك بالأرواح يا وطن.

- Motorola way of Encoding
  - Also known as Big Endian
    - Computer Words are stored such that Most Significant Bytes comes first

हालाँकि सूर के जीवन के बारे में कई जनश्रुतियाँ प्रचलित हैं, पर इन में कितनी सच्चाई है यह कहना कठिन है। कहा जाता है उनका जन्म सन् १४७८ में दिल्ली के पास एक ग़रीब ब्राह्मीण परिवार में हुआ। जनश्रुति के अनुसार सूरदास जन्म से ही अंधे थे। आजकल थी अंधे आदमी अक्सर 'सूरदास' कहलाते हैं। कई लोगों ने उन्हें गुरु के रूप में अपनाया और उनकी पूजा करना शुरु कर दिया ।

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Transfer Syntax

यदा यदा हि धर्मस्य ग्लानिर्भवति भारत ।
अभ्युत्थानमधर्मस्य तदात्मानं सृजाम्यहम् ॥ ७ ॥

Whenever and wherever a decline of righteousness and a predominance of unrighteousness prevails; at that time I manifest Myself personally, O descendant of Bhārata.

ಭರತ ವಂಶಜನಾದ ಅರ್ಜುನನೆ, ಯಾವಾಗ ಎಲ್ಲೆಲ್ಲಿ ಧರ್ಮದ
ಅವನತಿಯಾಗುತ್ತದೆಯೋ ಮತ್ತು ಅಧರ್ಮವು ಹೆಚ್ಚುತ್ತದೆಯೋ ಆಗ ಸ್ವಯಂ
ಅವತಾರ ಮಾಡುತ್ತೇನೆ

Transfer Syntax

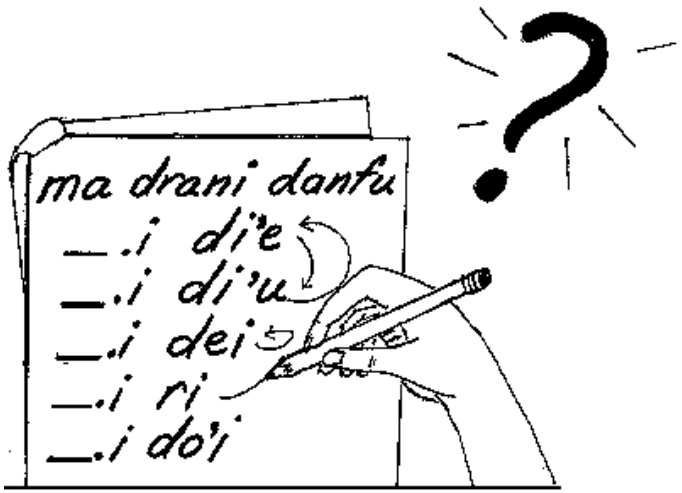
- The Format of Data Encoding

- Transfer Syntax Contains

  - *BYTE ORDERING*
    - Little Endian
    - Big Endian

  - *VALUE REPRESENTATION*
    - Implicit VR
    - Explicit VR

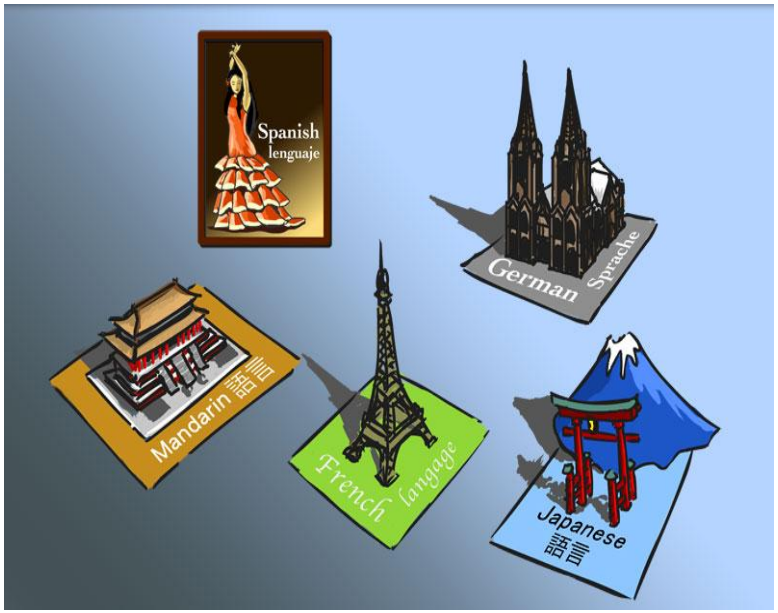
  - *COMPRESSION*
    - JPEG
    - RLE



Transfer Syntax

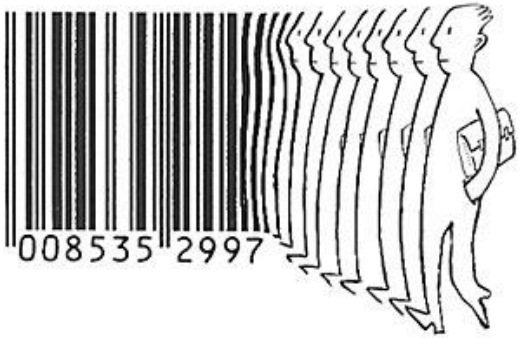
Few Transfer Syntaxes supported in DICOM

- Implicit VR Little Endian ← DICOM Default Transfer Syntax
- Explicit VR Little Endian
- Explicit VR Big Endian
- JPEG Lossless
- JPEG 2000
- RLE Lossless



Unique Identifiers

- IODs, Transfer Syntax are identified by Unique Identifiers (UID)
  - Identifier of an IOD is SOP Class UID
  - Identifier of Transfer Syntax is Transfer Syntax UID
- Universally Unique
- Assigned by NEMA



Unique Identifiers

UID Format

1.2.840.10008.XXXX

Where

1	ISO Organization
1.2	ISO Member Body
1.2.840	USA Branch of Member Body
1.2.840.10008	DICOM Standard

Similarly

1.2.840.10003	ANSI Standard Z.39
1.2.840.10006	IEEE 802.3

DICOM
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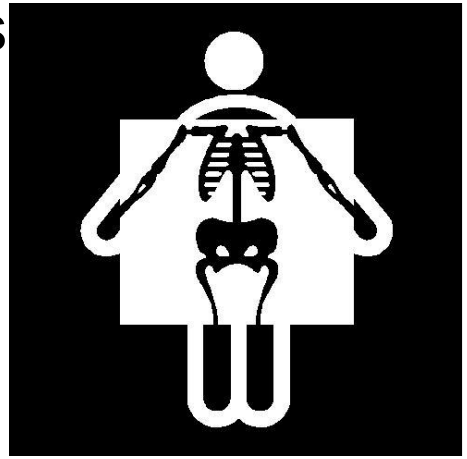
Unique Identifiers

- Image Identifiers are known as SOP Instance UID
- SOP Instance UID
  - Globally Unique
  - NEMA assigns prefix to Vendors
  - Suffix are added by Vendors for Various
  - Philips Prefix

1.3.46.670589



DICOM Data Encoding

- Demo
  - Patient Name/Patient ID
  - UIDs
  - Dump of DICOM Data

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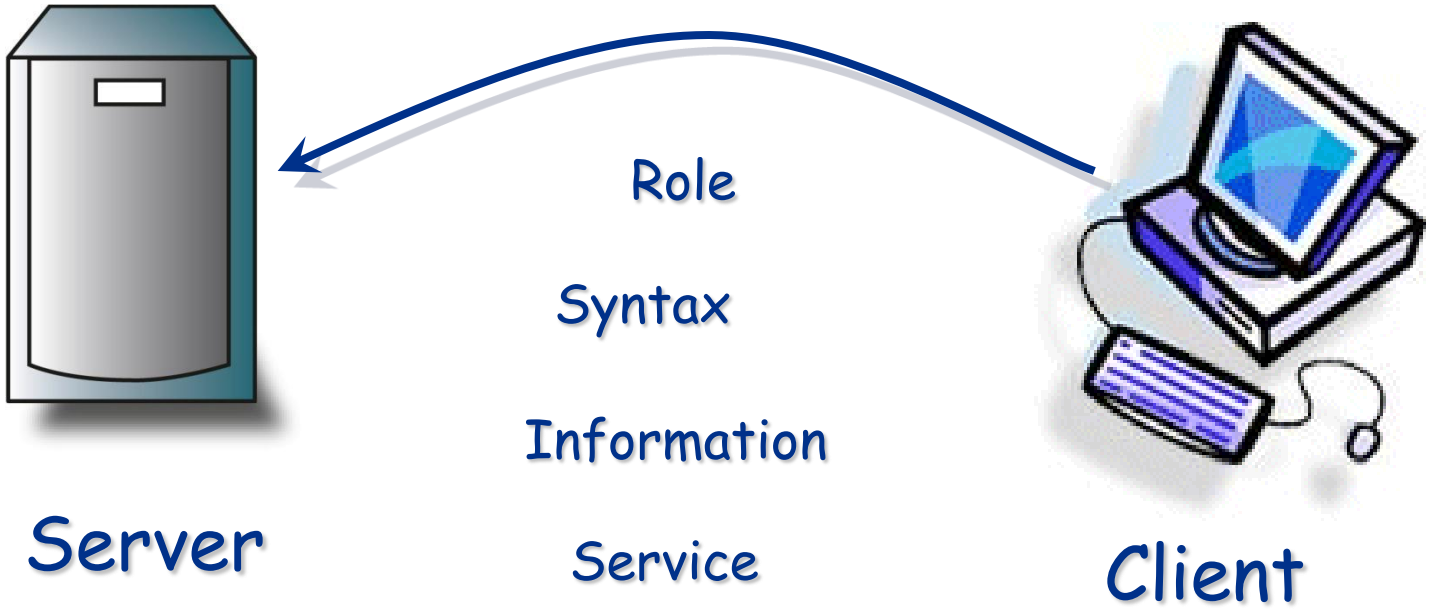
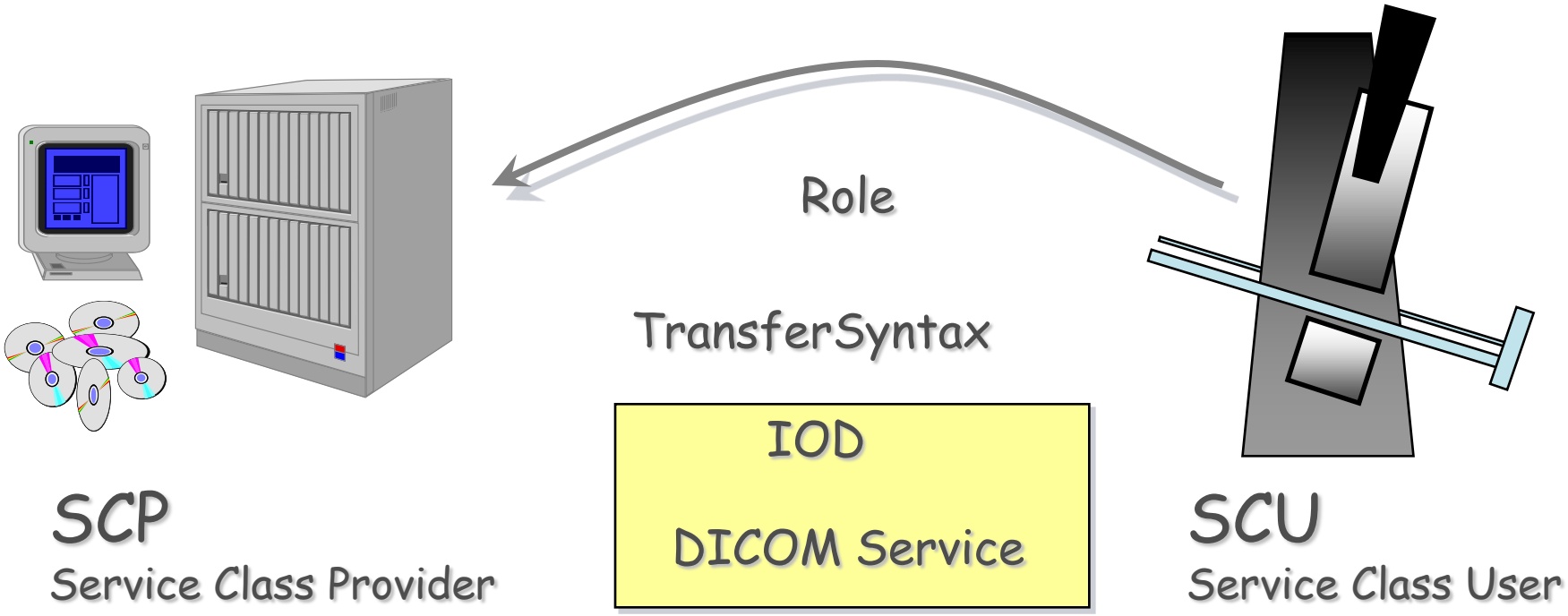
Client Server Model

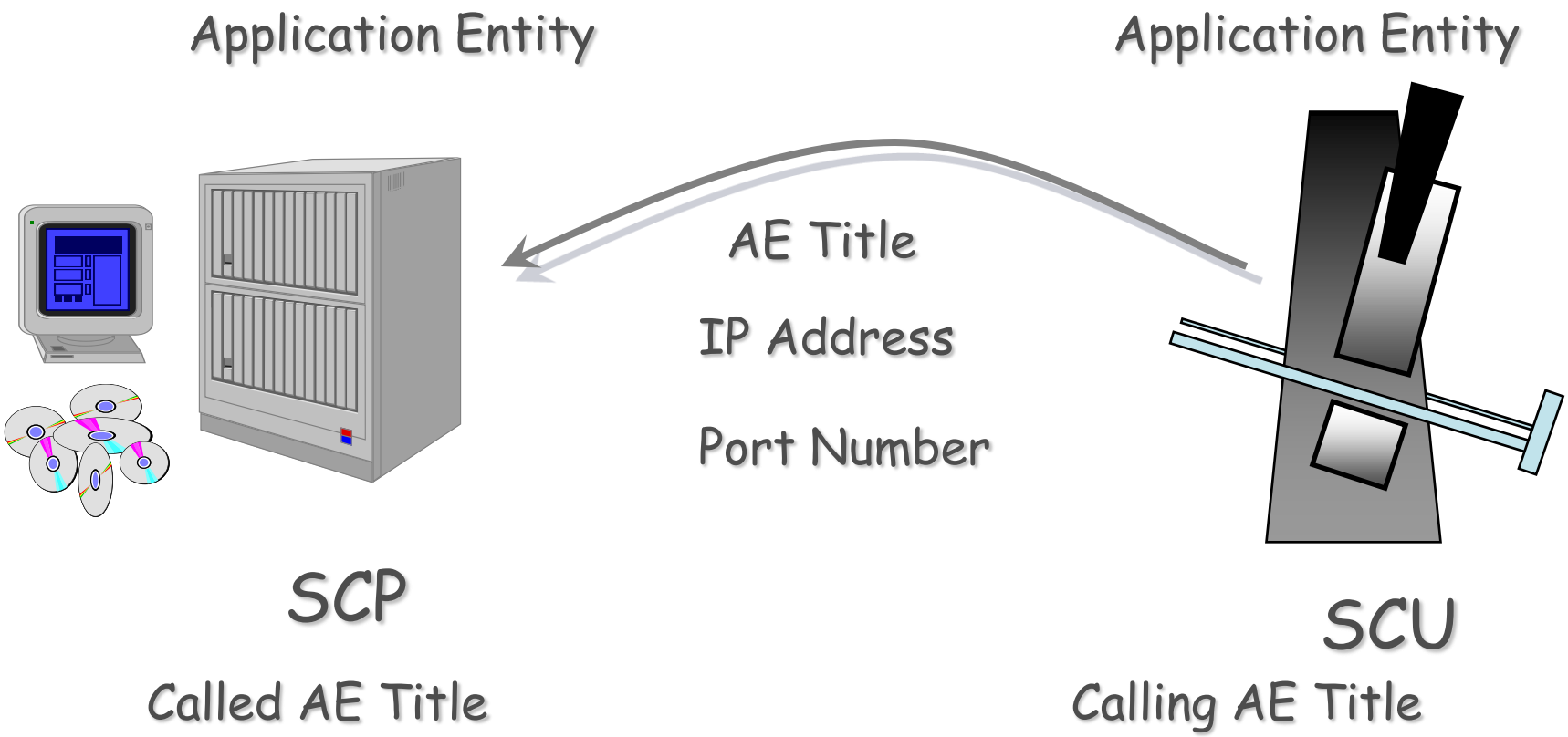


DICOM Network Model



Service Object Pair (SOP)

DICOM Communication



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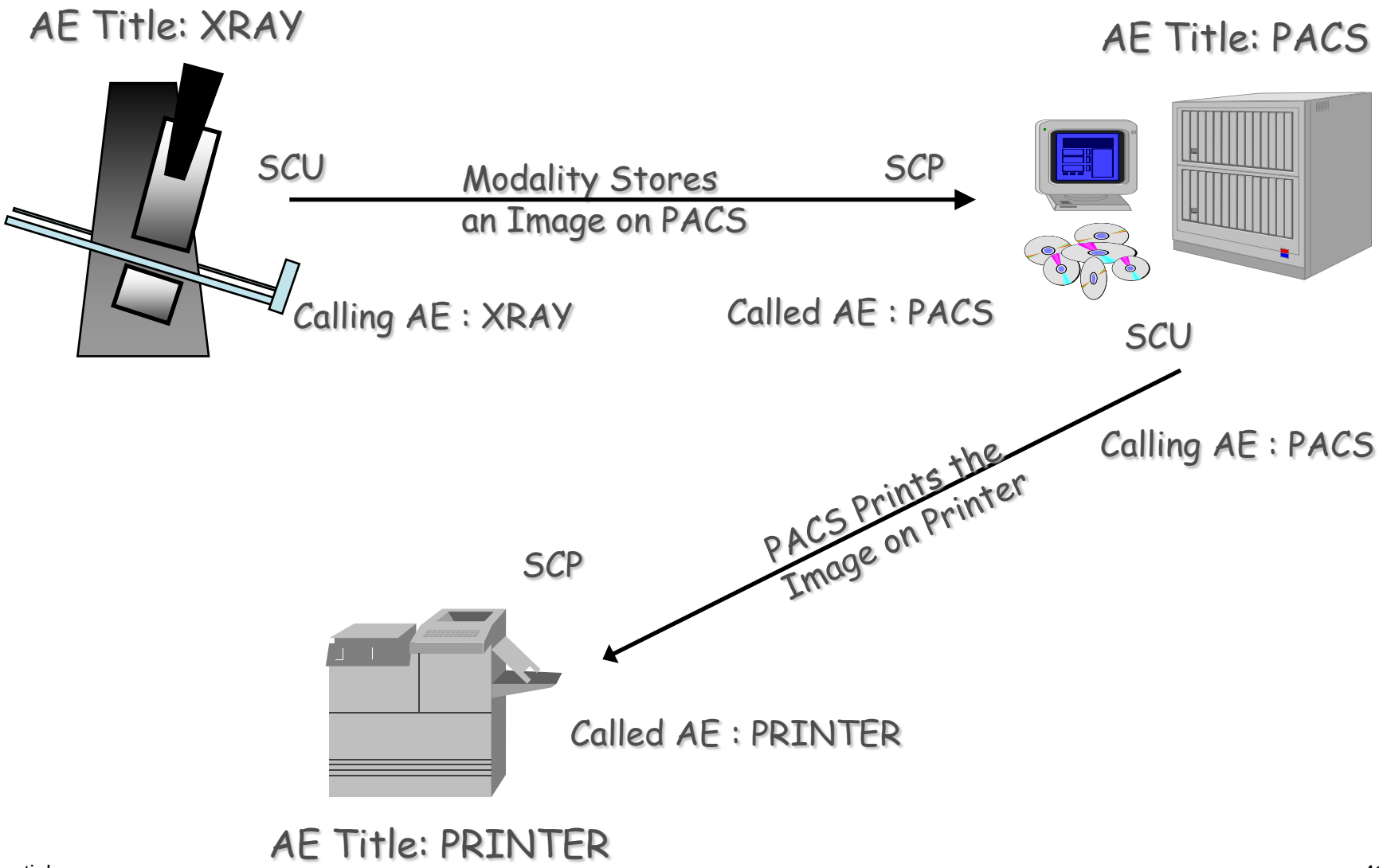
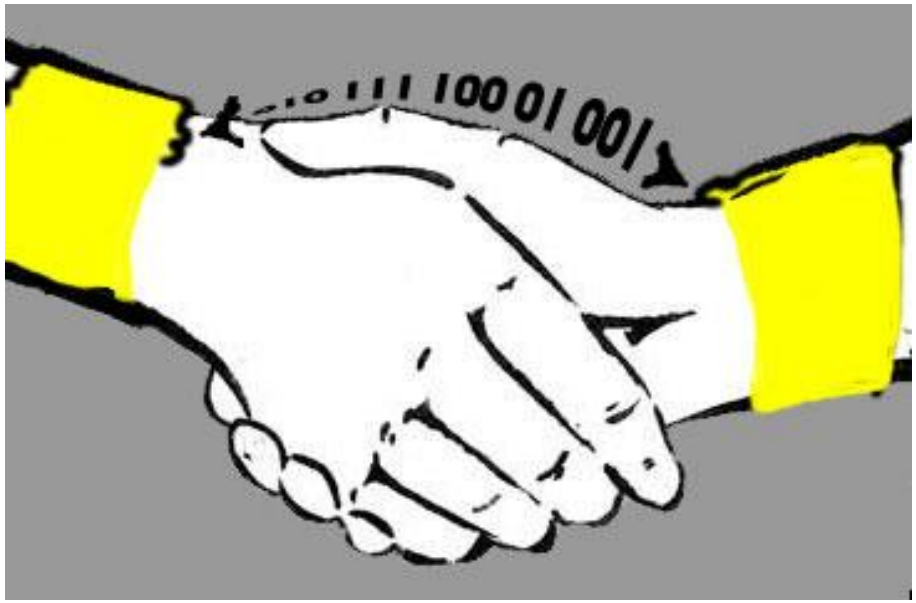
DICOM Network Model



DICOM Network Model

DICOM Handshake



Known as

DICOM Association

DICOM Association

- DICOM Association
  - SCU Initiates
  - SCP Accepts

- Association Message Contains
  - Called AE Title
  - Calling AE Title
  - IP Address & Port Numbers
  - One or more Presentation Contexts

DICOM Association

Presentation Context = Abstract Syntax + Transfer Syntax

Where
Abstract Syntax is the SOP Class for the Service
Transfer Syntax is the Format of Data Encoding

e.g: Storage of an MR Image in Implicit VR Little Endian

Where:
Abstract Syntax = MR Image Storage
Transfer Syntax = Implicit VR Little Endian

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Presentation Context

SCU Proposes

- One Abstract Syntax
- One or more Transfer Syntaxes for each Abstract Syntax

Presentation Context 1	Abs Syn MR Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	TS: JPEG Lossless

Presentation Context 3	Abs Syn US Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	TS: RLE Lossless

Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian

Presentation Context 7	Abs Syn Grayscale Print	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian

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Presentation Context

- SCP Accepts
  - One Abstract Syntax
  - One Transfer Syntax

- SCP can also Reject one or more Presentation Contexts

Presentation Context 1	Abs Syn MR Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	TS: JPEG Lossless

~~Presentation Context 3~~	Abs Syn US Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	TS: RLE Lossless

Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian

~~Presentation Context 7~~	Abs Syn Grayscale Print	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian

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Presentation Context

Presentation Context 1	Abs Syn MR Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	TS: JPEG Lossless
Presentation Context 3	Abs Syn US Image Storage	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	TS: RLE Lossless
Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Little Endian	TS: Explicit VR Big Endian	
Presentation Context 7	Abs Syn Grayscale Print	TS: Implicit VR Little Endian	TS: Explicit VR Little Endian	

Presentation Contexts Proposed

Presentation Context 1	Abs Syn MR Image Storage	TS: Explicit VR Big Endian
Presentation Context 5	Abs Syn Query for Patient	TS: Implicit VR Big Endian

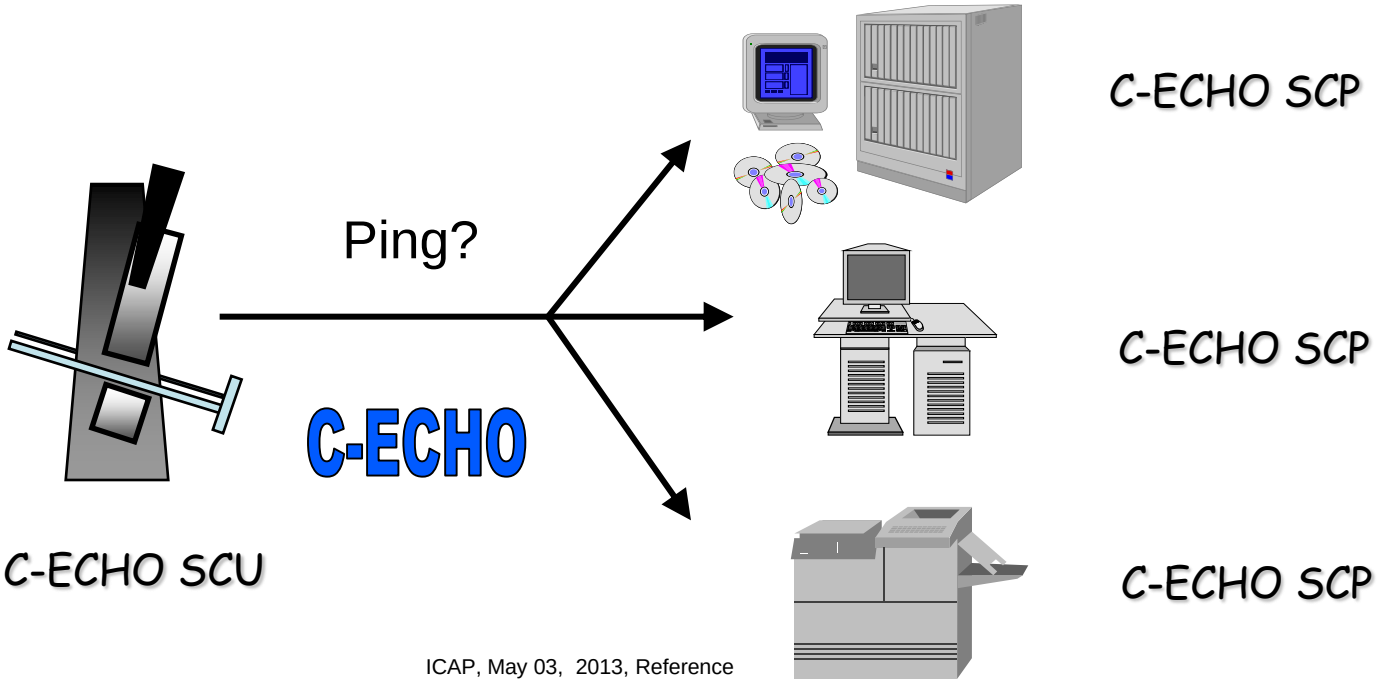
Presentation Contexts Accepted

DICOM Network Service

- Two Types of Network Services
  - Composite Services
    - Services Related to Persistent objects
      - Verification
      - Storage
      - Query/Retrieve
      - Modality Worklist
  - Normalized Services
    - Services Related to Transient information
      - Storage Commitment
      - MPPS
      - Print

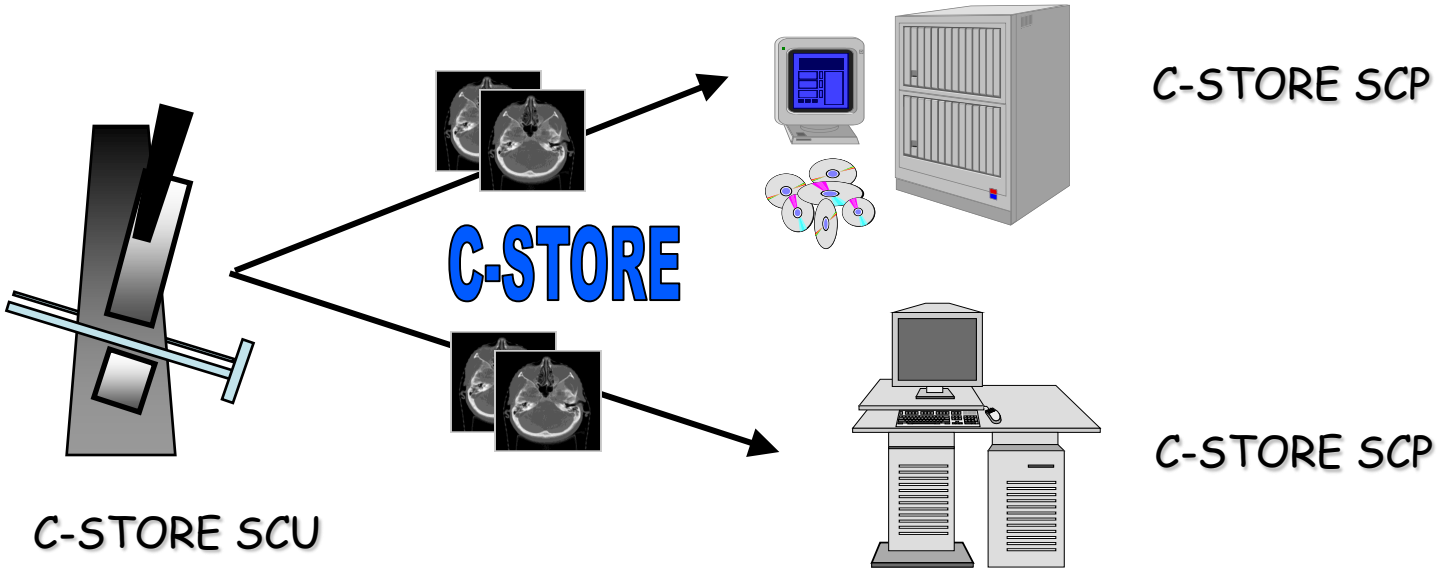
Verification Service

- Verifies the Availability of the Remote Node
- Implemented using the command DIMSE **C-ECHO**



Storage Service

- Transfers Composite Instances to a Remote System
- Used by Modalities/Work Stations to Save Images on PACS
- Implemented using the command DIMSE **C-STORE**



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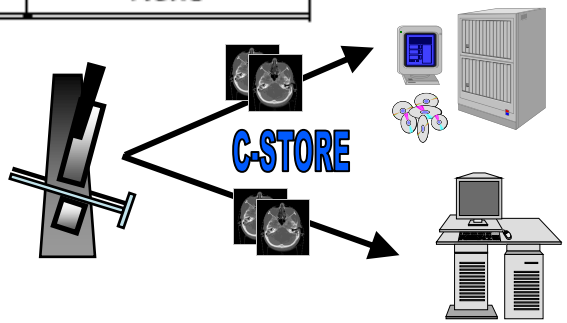
Storage Service

C-STORE Status Codes

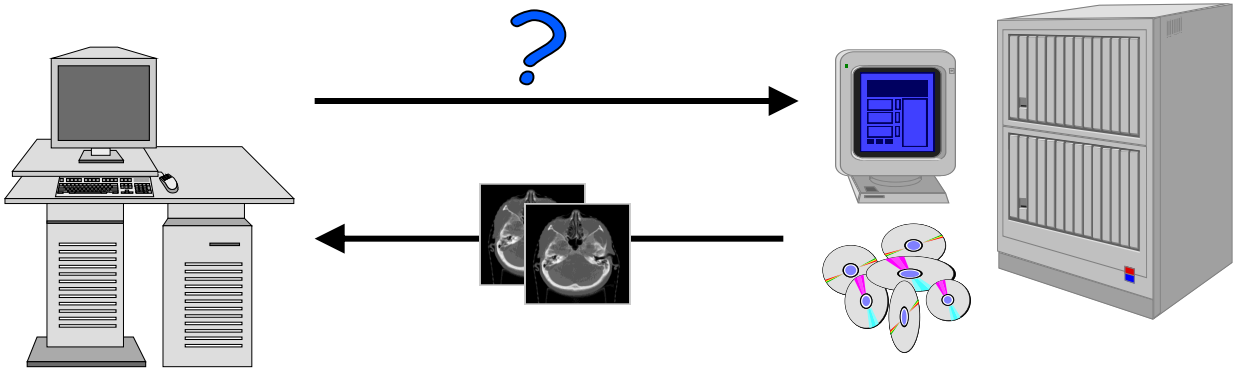
Service Status	Further Meaning	Status Codes	Related Fields
Failure	Refused: Out of Resources	A7xx	(0000,0902)
	Error: Data Set does not match SOP Class	A9xx	(0000,0901) (0000,0902)
	Error: Cannot understand	Cxxx	(0000,0901) (0000,0902)
Warning	Coercion of Data Elements	B000	(0000,0901) (0000,0902)
	Data Set does not match SOP Class	B007	(0000,0901) (0000,0902)
	Elements Discarded	B006	(0000,0901) (0000,0902)
Success		0000	None

Table from DICOM 3.4.B.2.3



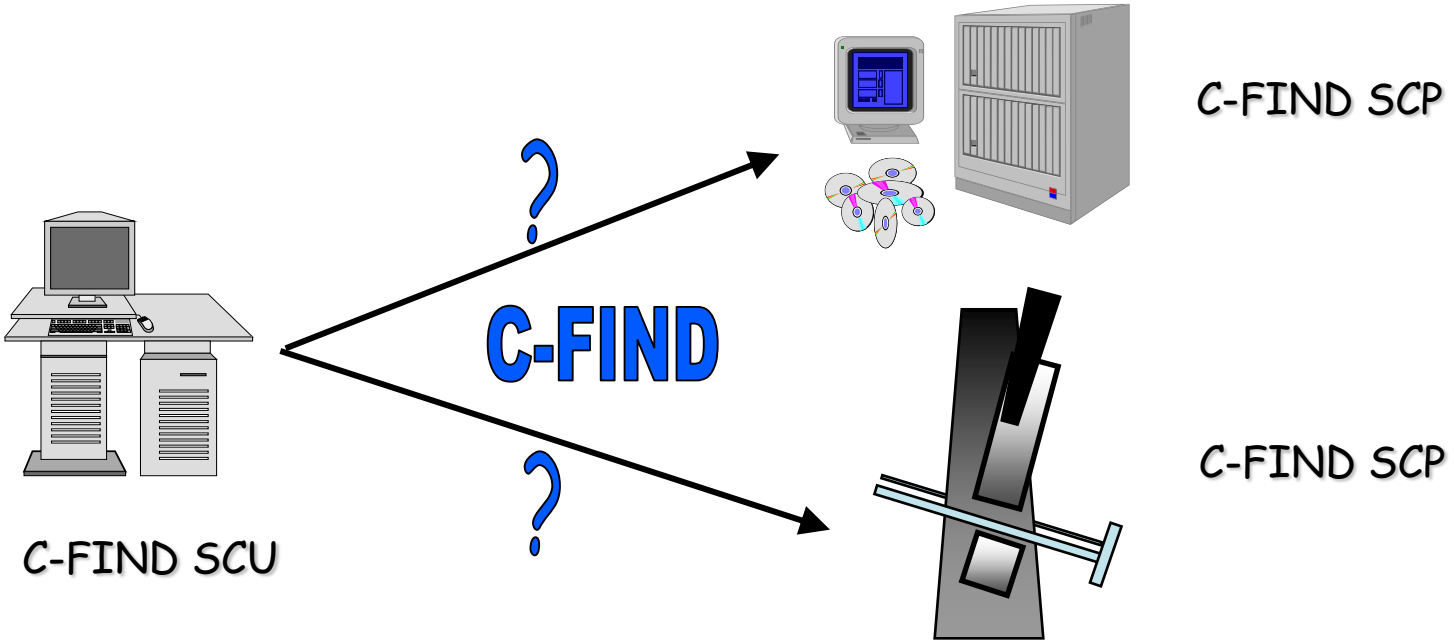
Query/Retrieve Service

- Used For
  - Locating the Objects on a Remote System
  - Retrieving the Relevant objects from a Remote System
- Used by Work Stations to Retrieve the Objects from PACS/Modalities
- Implemented by DIMSE **C-FIND & C-MOVE**



Query/Retrieve : C-FIND

- Query the Remote System for Composite Objects
- Used by Work Stations to Query for the Objects on PACS/Modalities



PHILIPS

Query/Retrieve : C-FIND

- Attribute Matching
  - Single Value Matching
    e.g: Patient ID == “1234”
  - Wild Card Matching
    e.g: Patient Name == “A*”, Patient ID == “123?”
  - Universal Matching
    e.g: Patient Name == “” / “*”
  - Range Matching
    e.g: Study Date >= “20070701”
    Study Time <= “153020”
    Study Date >=“20070701” &
    Study Date <= “20070723”

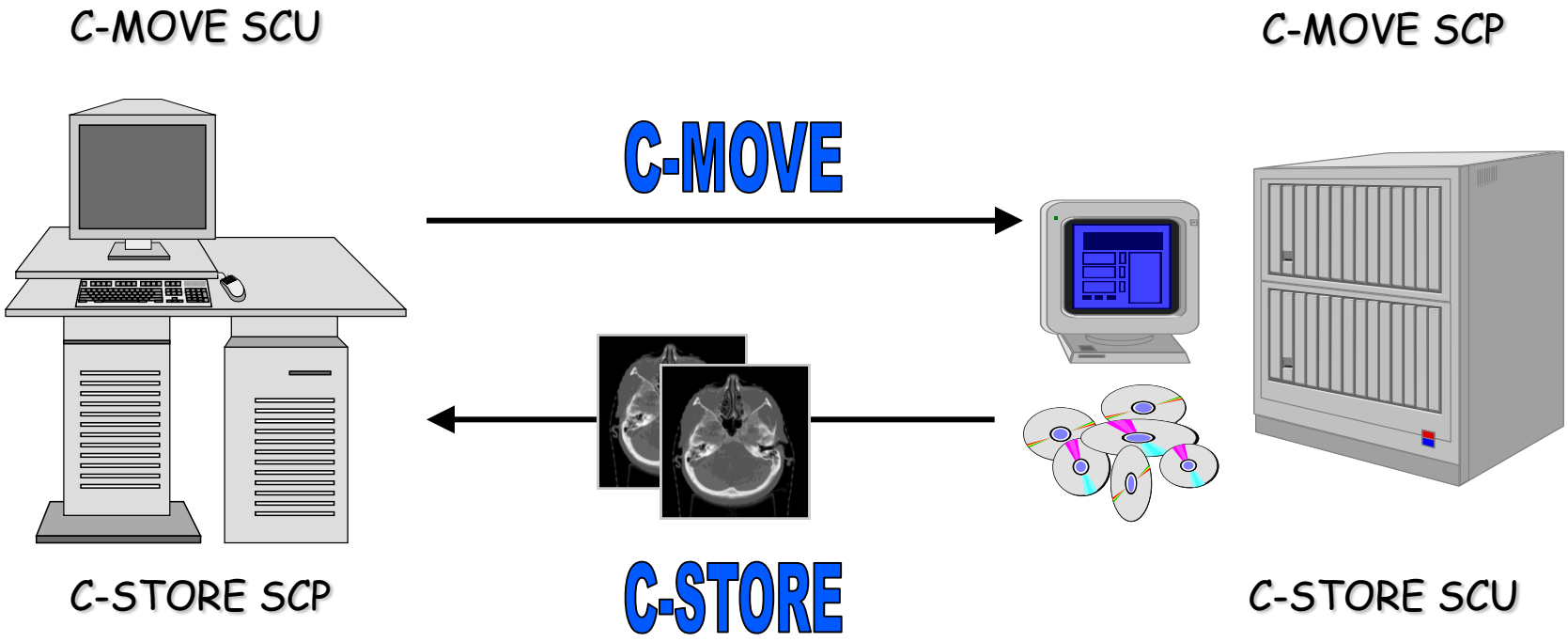
PHILIPS

Query/Retrieve : C-MOVE

- Transfers Composite Objects from One Remote System to another
- Used by Work Stations to Retrieve the Objects from PACS/Modalities
- A C-MOVE always results in a C-STORE
- C-MOVE always results in 2 DICOM Associations
  - One for C-MOVE
  - One for C-STORE
- C-MOVE can be realized as Two Ways

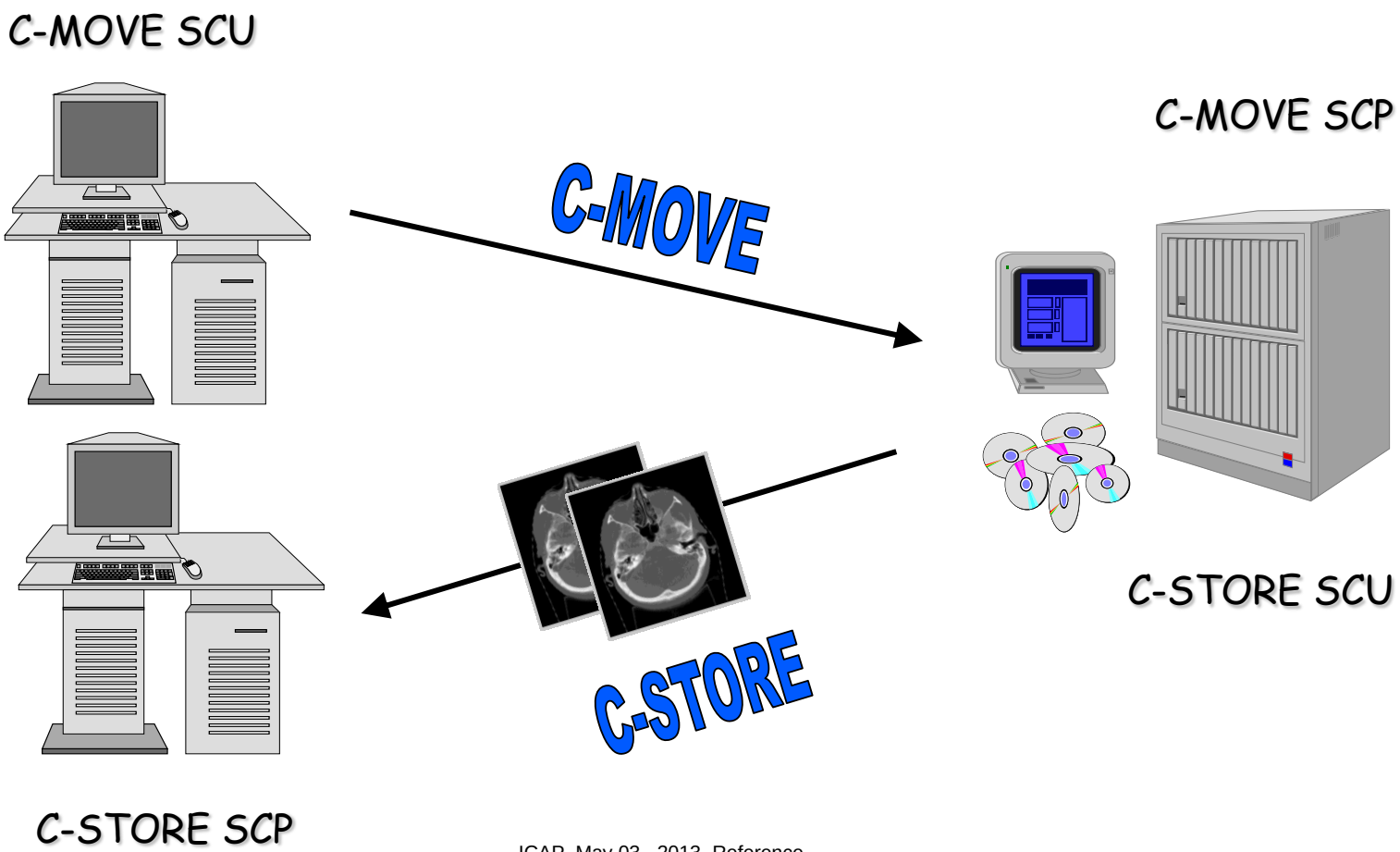
Query/Retrieve : C-MOVE

1. A Workstation Retrieving the Images from PACS



Query/Retrieve : C-MOVE

2. A Workstation Transferring the Images from PACS To another Workstation



Normalized Services

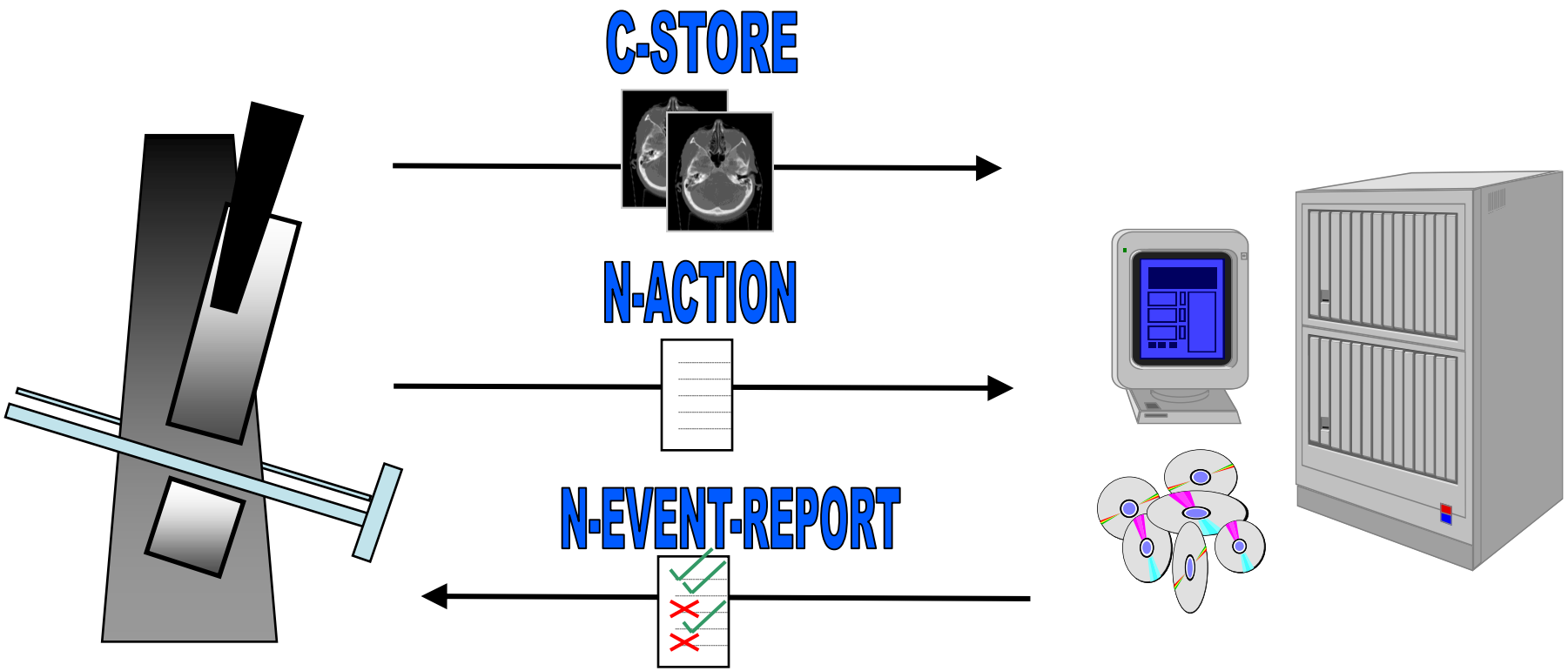
PHILIPS

Storage Commitment

- Storage Service
  - Allows SCU to Transmit Images to SCP
  - SCP may delete the Data
  - Does not explicitly ensure the Safekeeping of received Data
- Need for Explicit Commitment of the Stored Objects
- Storage Commitment
  - Transfers the Responsibility of Objects to the SCP
  - SCP Implementation defines the Safekeeping Policy
  - Used by Modalities to make sure that they can delete the Composite Objects

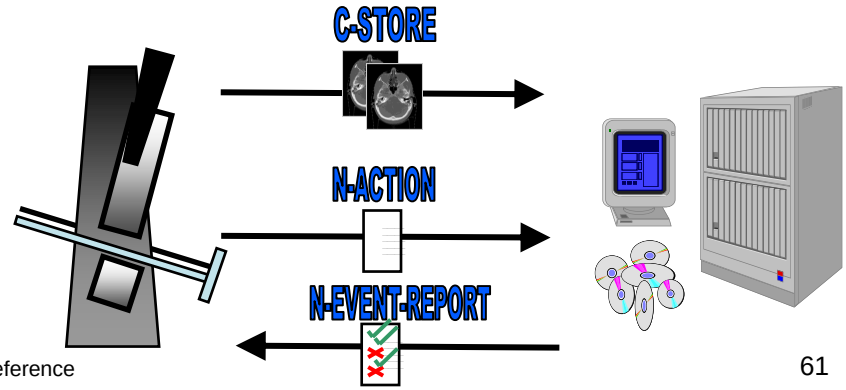
Storage Commitment

- Implemented using the Storage Commit SOP Class
- Implemented by DIMSE **N-ACTION & N-EVENT-REPORT**



Storage Commitment

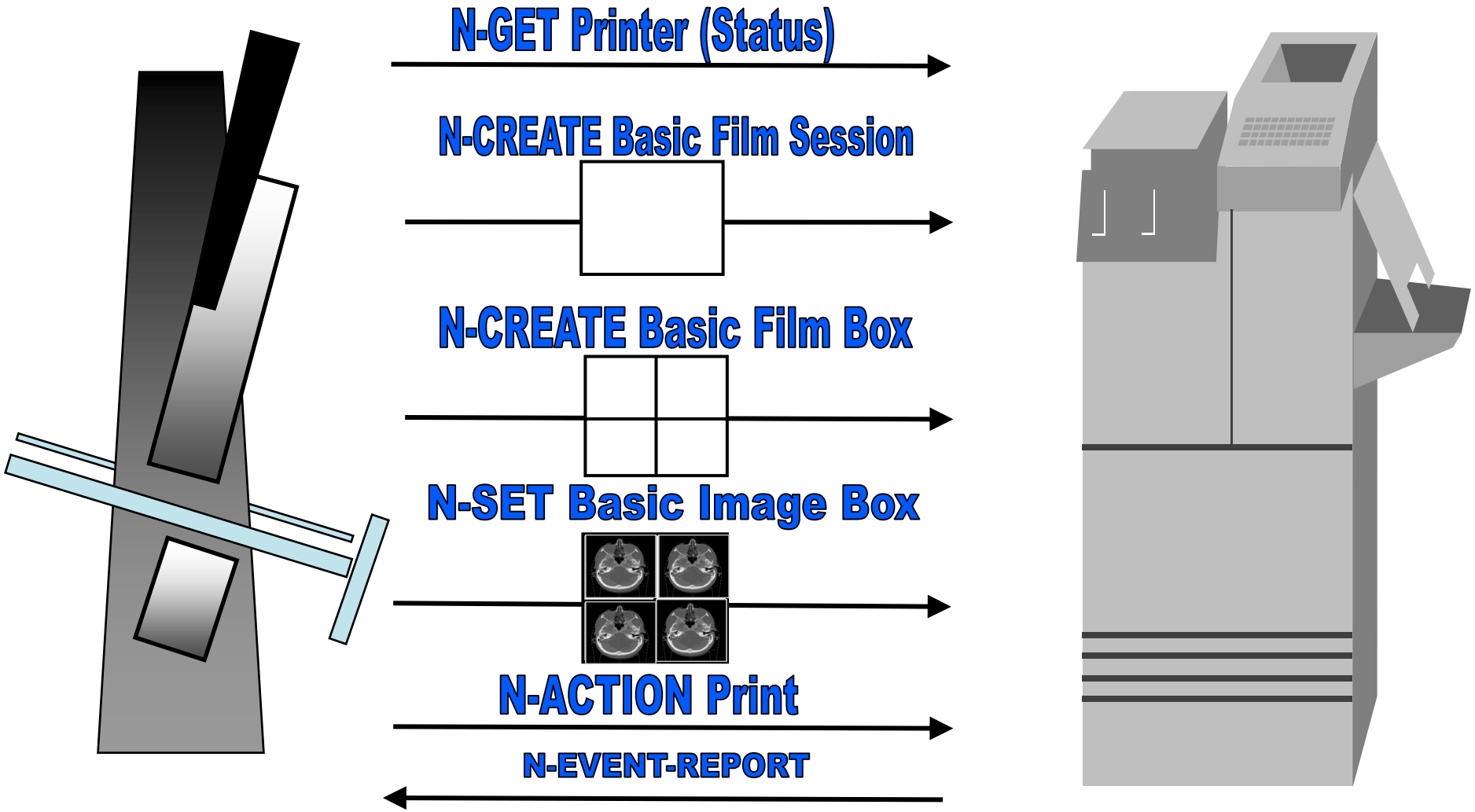
- Storage Commit Request (N-ACTION) Contains
  - Transaction UID
  - List of SOP Class UID & SOP Instance UID
- Storage Commit Response (N-EVENT-REPORT) contains
  - Transaction UID
  - Successful SOP Class UID & SOP Instance UID
  - Failed SOP Class UID & SOP Instance UID and Reason for Failure



Print Management

- Prints the Images on a Film
- Implemented by Basic Grayscale/Color Print Management Meta SOP Class
  - Meta SOP Class is used only for Association
  - Needs Four SOP Classes to perform the Print
    - Basic Film Session SOP Class
    - Basic Film Box SOP Class
    - Basic Grayscale/Color Image Box SOP Class
    - Printer SOP Class
  - Optionally Printer Job SOP Class

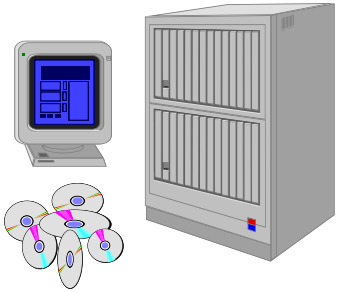
Print Management Sequence

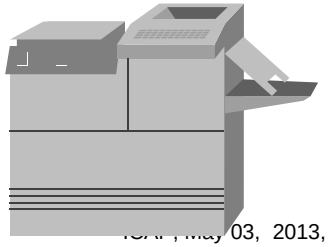


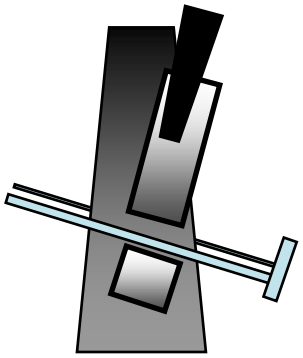
PHILIPS

Exercise

Identify the Order of DICOM Communication along with the DICOM Commands and Roles possible.







PHILIPS

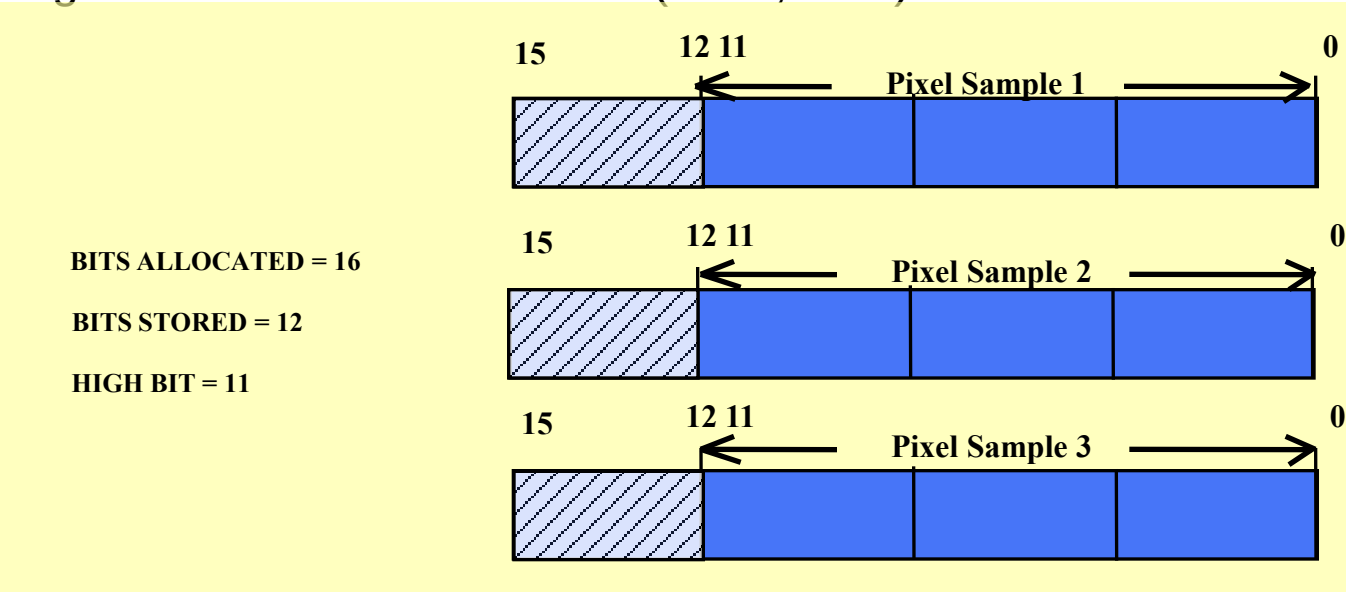
Agenda

- Introduction to DICOM
- Information Model
- Data Encoding
- Network Model
- Image & Presentation State
- DICOM Conformance

DICOM Pixel Data Encoding

- The Pixel Related Info is part of Image Pixel Module

- Properties of A Single Pixel
  - Bits Allocated (0028,0100)
  - Bits Stored (0028,0101)
  - High Bit (0028,0102)



PHILIPS

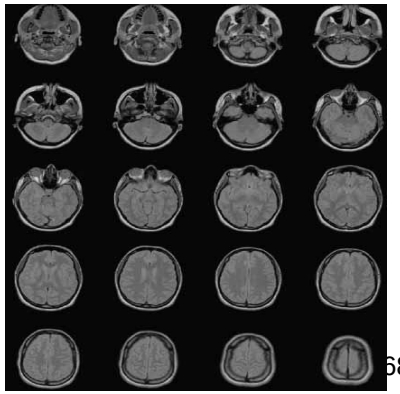
DICOM Pixel Data Encoding

- Major attributes that determine the Pixel Data Properties
  - Rows, Columns (0028,0010), (0028,0011)
    - Determine whether a Image is a Square or Non Square
  - Photometric Interpretation (0028,0004)
    - Monochrome1, Monochrome 2
    - Palette Color
    - RGB/YBR
  - Samples per Pixel (0028,0002)
    - Monochrome = 1 or Color = 3
  - Planar Configuration (0028,0006)
    - R1R2…Rn G1G2…Gn B1B2…Bn
    - R1G1B1 R2B2G2 … RnGnBn
  - Pixel Data (7FE0,0010)

PHILIPS

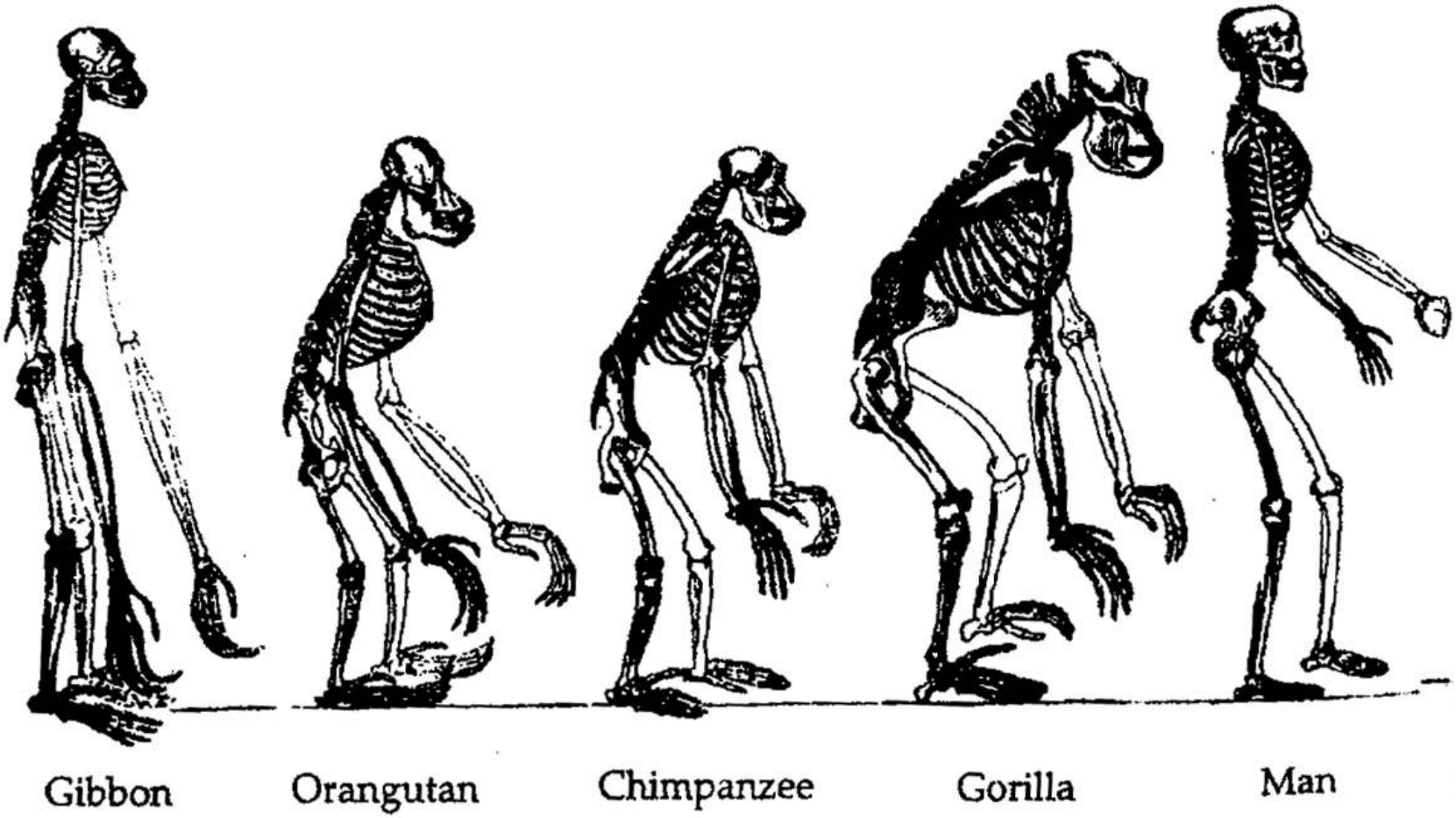
DICOM Pixel Data Encoding

- Multiframe Images
  - Images whose Pixel Data contains sequence of Individual Image Frames
  - Single Contiguous stream of Pixels and without Frame Headers
  - All Frames are of Equal Size.
  - Size of Pixel Data = Pixel Size for 1 Frame x Nr. Of Frames



PHILIPS

DICOM Pixel Data Transformation

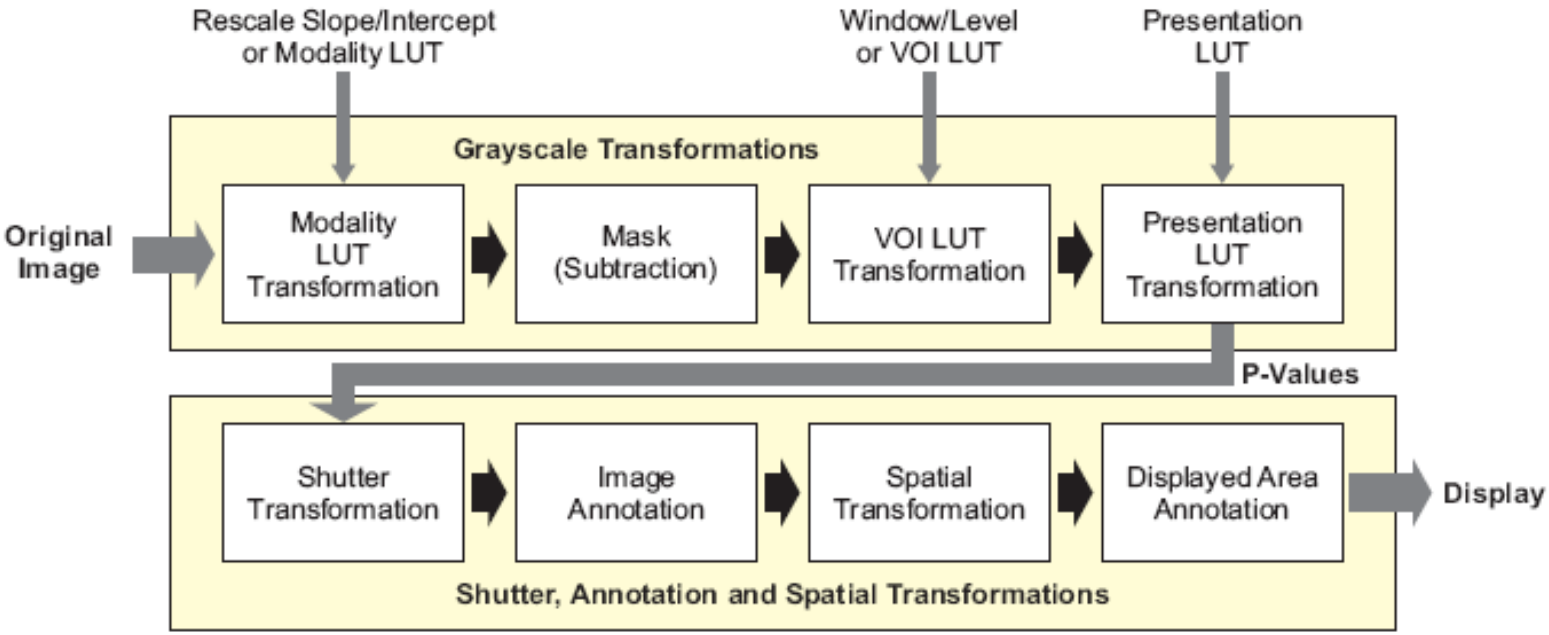


PHILIPS

DICOM Pixel Data Transformation

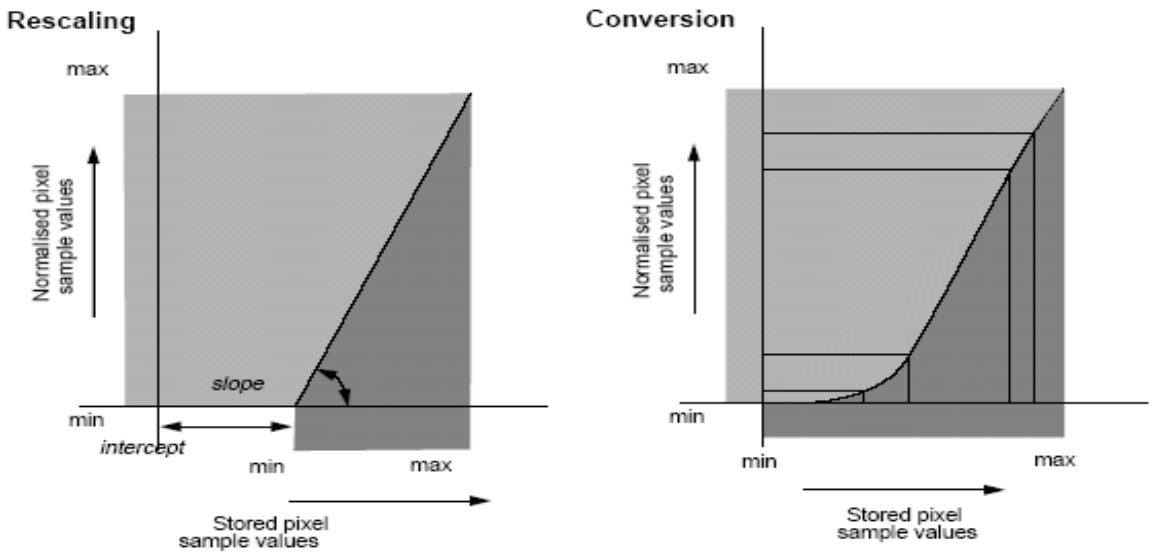
- Pixel Data undergoes a series of transformations before being rendered on screen

- Transformations are commonly specified in the Presentation State IOD



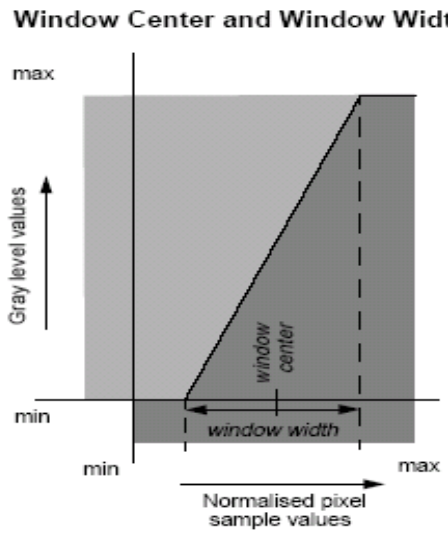
DICOM Pixel Data Transformation

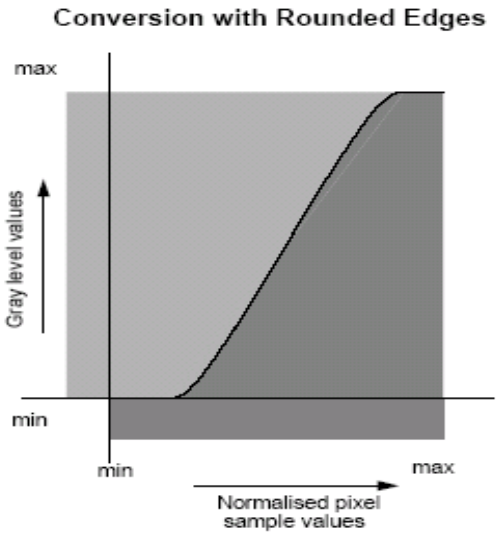
- Modality Look Up Table (LUT)
  - Manufacturer Dependent Pixel Values to Manufacturer independent values
  - Attributes: Rescale Slope & Rescale Intercept



DICOM Pixel Data Transformation

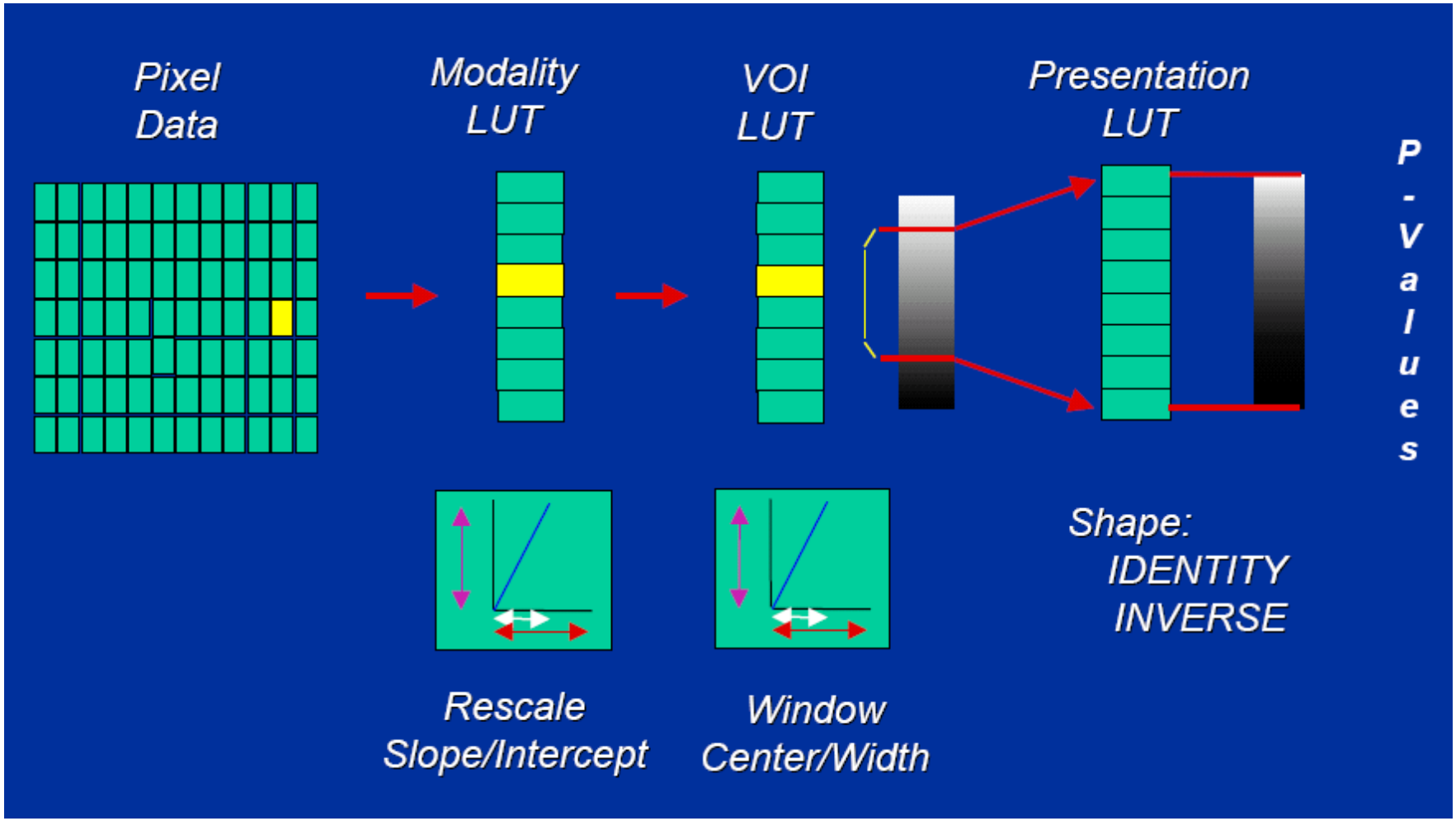
- Value of Interest (VOI) LUT
  - Scaling of selected Input values to Full Grayscale Range
  - Output of Mask is the Input for VOI LUT
  - Attributes : Window Centre & Window Width





PHILIPS

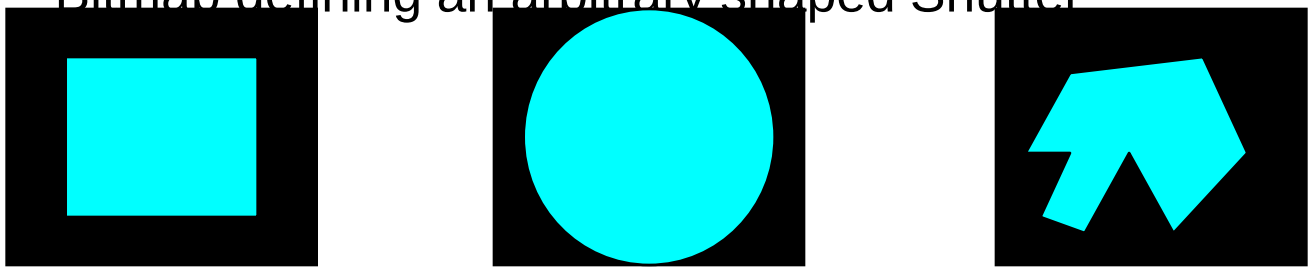
DICOM Pixel Data Transformation



Courtesy David Clunie, RadPharm

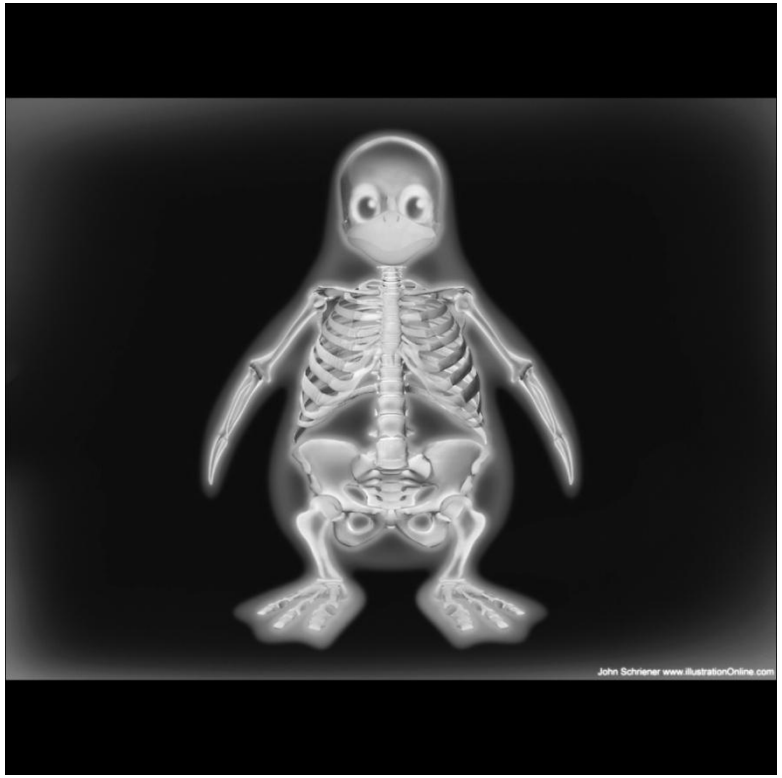
DICOM Pixel Data Transformation

- Shutter Transformation
  - Mask which may be applied on the image for presentation purposes
  - Neutralizes the display of pixels located outside the shutter
  - Display Shutter
    - Geometric shape expressed as Row to Columns Co-ordinates
  - Bitmap Shutter
    - Bitmap defining an arbitrary shaped Shutter



DICOM Pixel Data Transformation

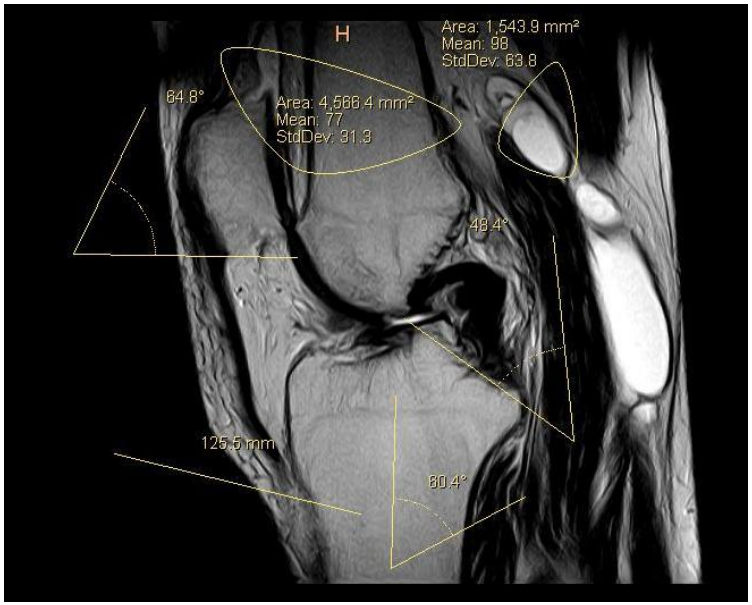
Shutter Transformation



PHILIPS

DICOM Pixel Data Transformation

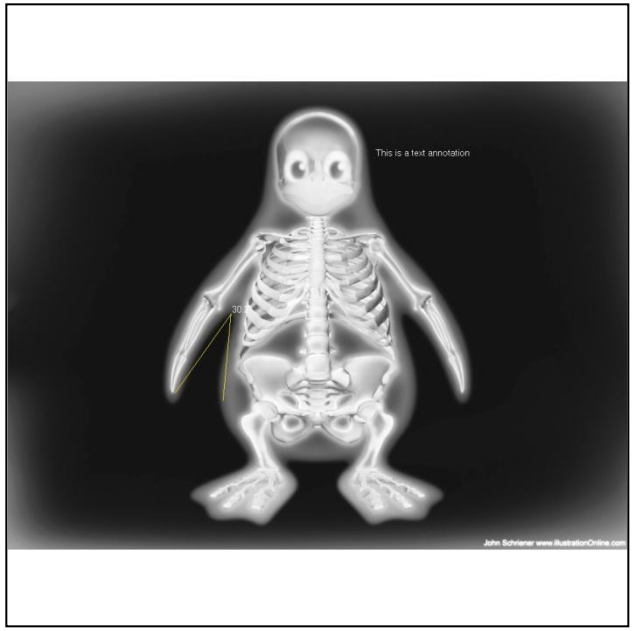
- Image Annotation
  - Vector Graphics and/or Text Annotations applied on the Image
  - Position and Size are relative to the Pixel Co-ordinates
  - ROI as Line, Angle, Circle, Ellipse etc and can be filled
  - May contain Textual annotations

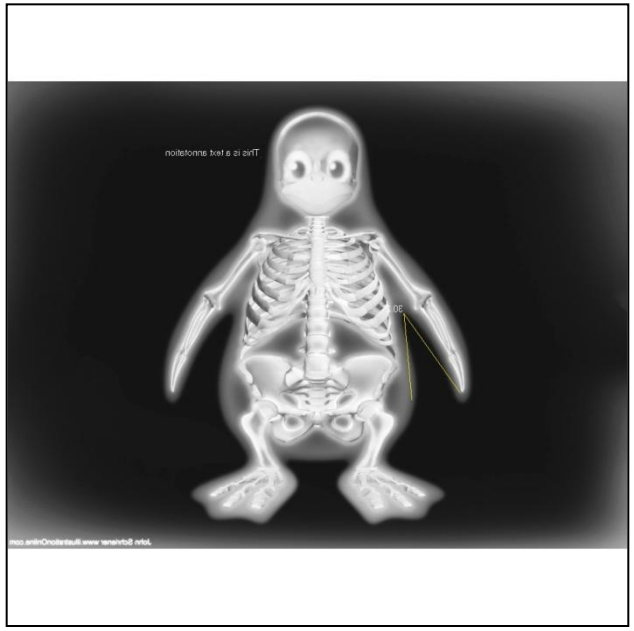


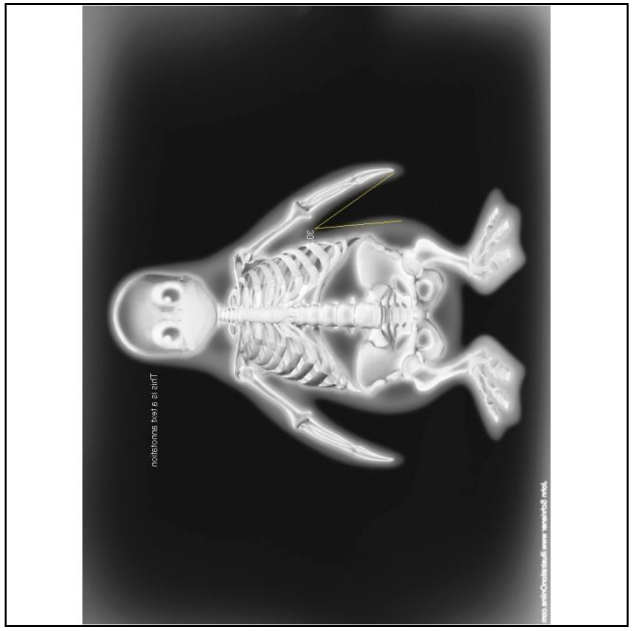
PHILIPS

DICOM Pixel Data Transformation

- Spatial Transformation
  - Describes the Rotation of the Image
  - May contain Horizontal Flipping after rotation

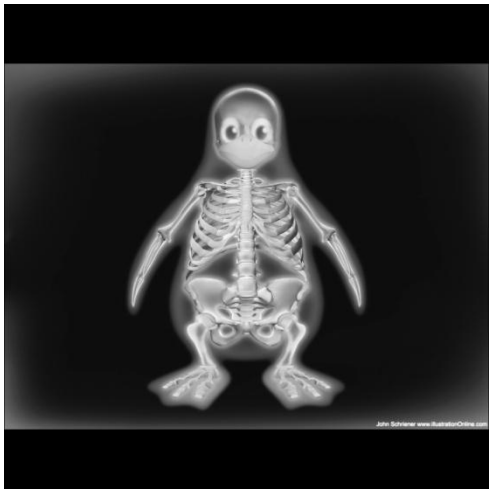


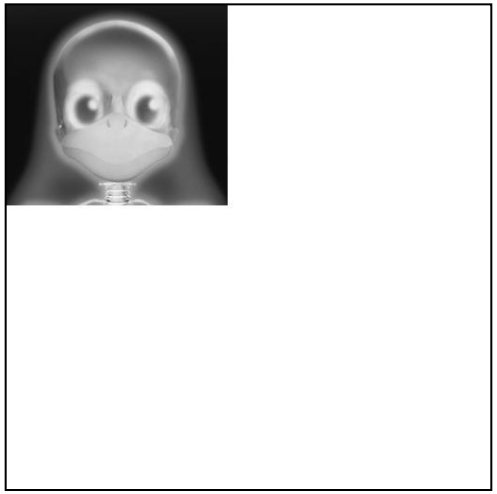


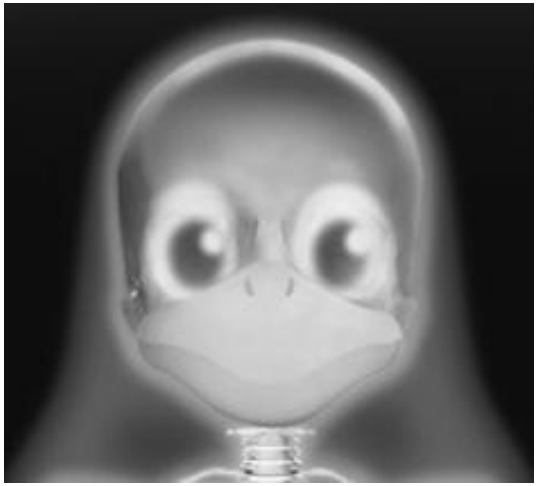


DICOM Pixel Data Transformation

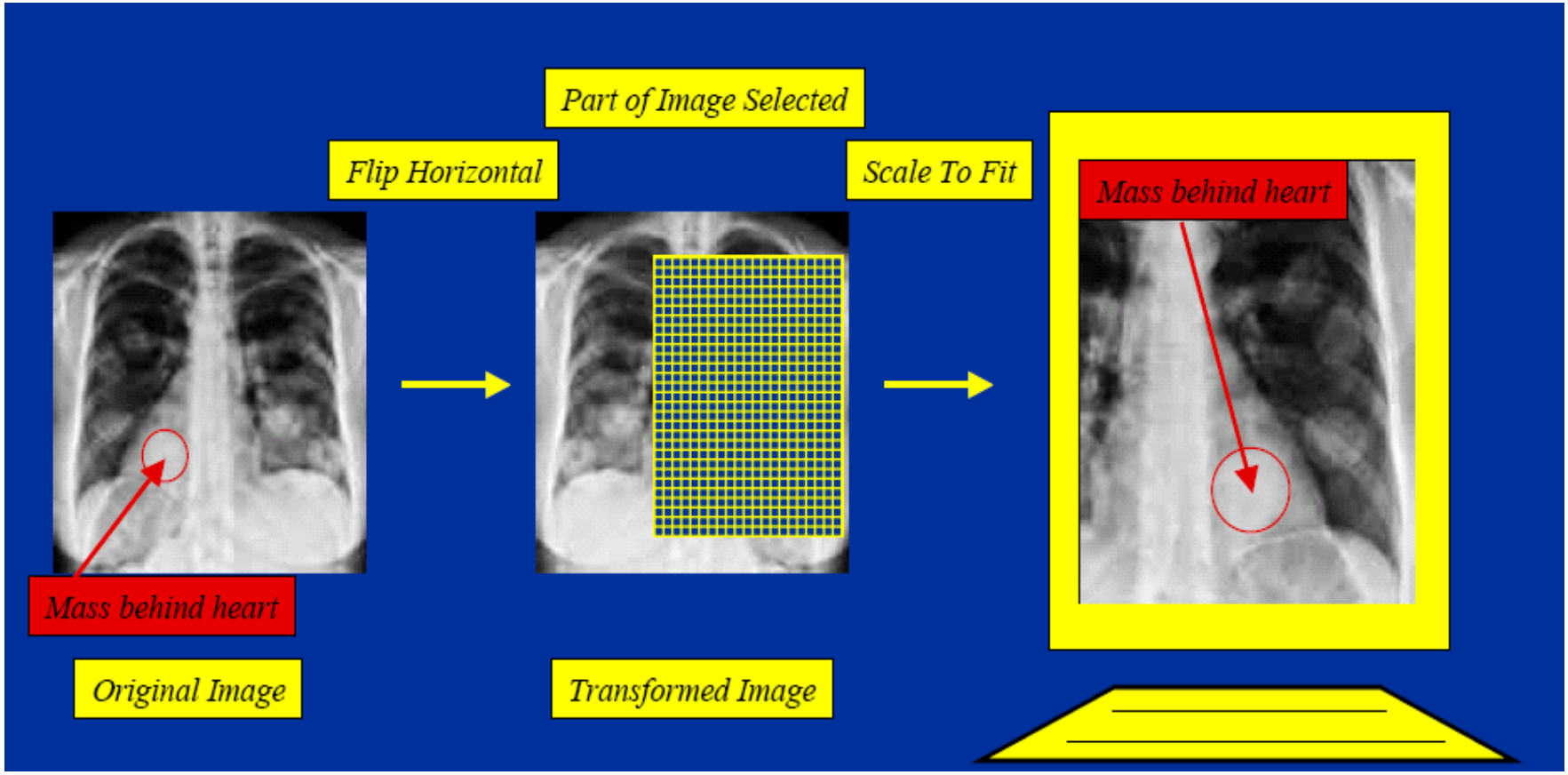
- Displayed Area Transformation
  - Displayed Area is the portion of the image displayed on the device
  - SCALE TO FIT, TRUE SIZE, MAGNIFY
  - Graphic Annotation Module are rendered relative to the Specified Displayed Area, not the actual rendered displayed area







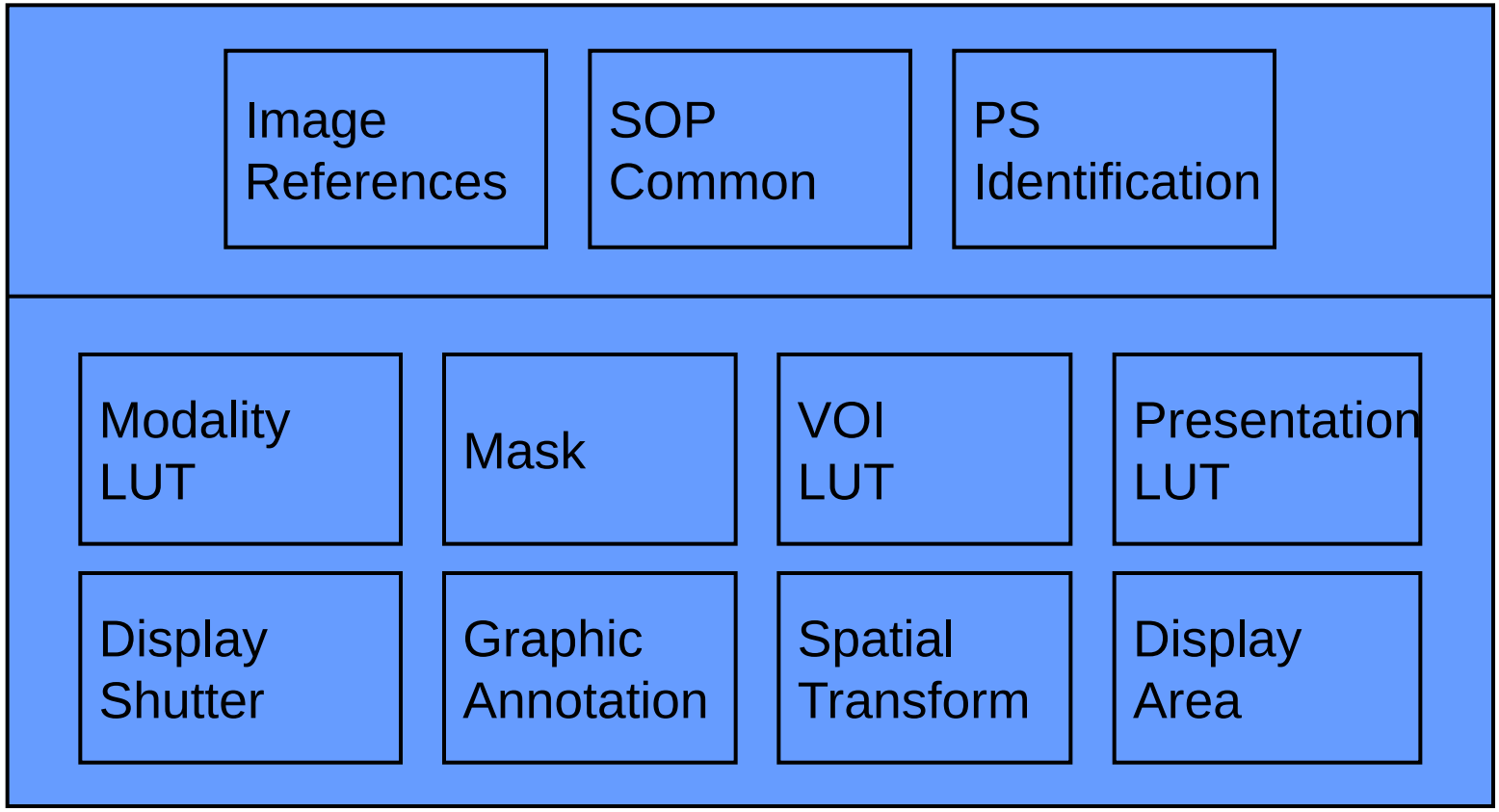
DICOM Pixel Data Transformation



Courtesy David Clunie, RadPharm

PHILIPS

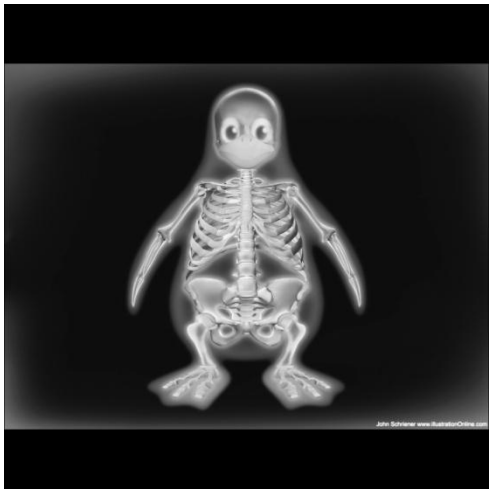
Presentation State

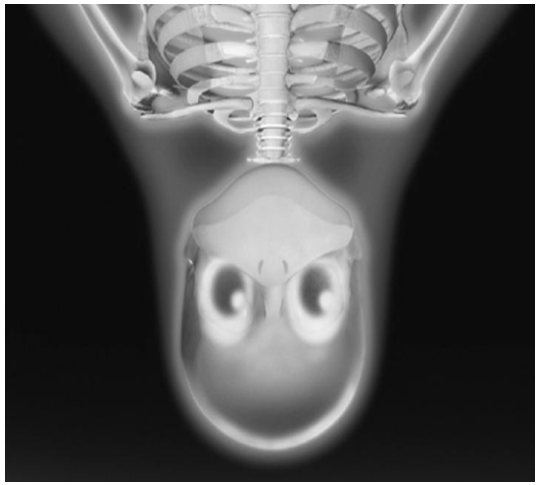


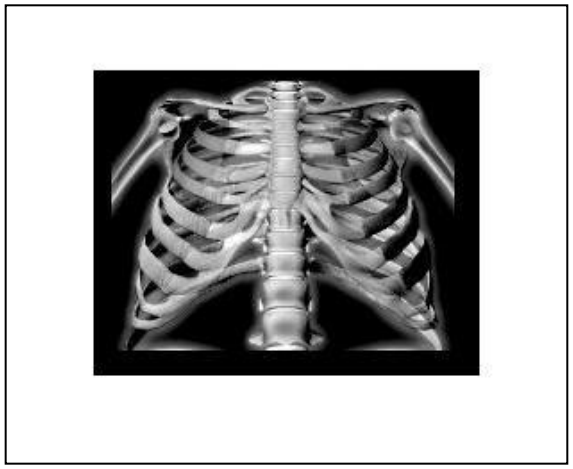
PHILIPS

Presentation State

- Separation of Stored Image Instances from Display characteristics
- Contains references to one or more Images
- One Image can have multiple Presentation States
- Stored in a Separate Series with Modality as “PR”







PHILIPS

Agenda

- Introduction to DICOM
- Information Model
- Data Encoding
- Network Model
- Image & Presentation State
- DICOM Conformance

PHILIPS

DICOM Conformance

- DICOM does not
  - Validate the Conformance of Products
  - Provide Certification for Products
  - Point fingers at Products
  - Do Policing

- Why DICOM Conformance
  - Required by the Law
  - To Identify the Capabilities of System
  - To Deduce Compatibility between Systems
  - To Identify Interoperability in a Healthcare Enterprise

PHILIPS

DICOM Conformance Statement

- Follows a Standard structure

- Conveys the DICOM Features implemented
  - Information Object Definitions Supported
  - Services Classes Provided and their Roles

- DICOM Conformance Statement Helps
  - In Identifying Interoperability between products
  - Hospital take fast decision on products
  - Service Engineers to setup the System
  - To Resolve issues to Avoid Finger Pointing

PHILIPS